

CCPP Piping Guide Guidance Book

*Combined Cycle Power Plant
Piping Construction Guidebook*

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1. Purpose

- 1.1. This guide should be used as a practical reference for combined cycle power plant piping construction. By adhering to these principles and prioritizing safety and quality at every stage, it will lead to the successful and reliable operation of the entire power plant. Please note that some content may be subject to change based on the project specification, application code, and design criteria.

2. Piping Construction in a CCPP Project

- 2.1. In a CCPP project, piping works are generally carried out in the sequence of utility supply and operating facilities, followed by the Simple Cycle stage (GT operation), and then the Combined Cycle stage (HRSG and Steam Generator operation).

2.2. Utility System

- 2.3.1. Utility systems of a combined cycle power plant form the essential service networks that support both construction and operational phases. These systems include service water, compressed air, demineralized water, fire water, nitrogen, and other auxiliary lines necessary for construction activities, commissioning, and plant start-up.
- 2.3.2. Their installation is prioritized during early construction, as they provide the basic infrastructure for piping works, equipment erection, and testing. The primary role of utility systems is to supply critical services such as water for cleaning and hydro testing, compressed air for pneumatic tools and valve actuation, demineralized water for flushing and pre-commissioning, and fire water for safety during hot works. By delivering these services, the systems ensure efficient project execution, protect equipment integrity, and safeguard personnel.
- 2.3.3. Early completion of utility systems establishes the foundation for safe, orderly, and uninterrupted progress throughout construction and commissioning.
- 2.3.4. **Firefighting system**
 - 2.3.4.1. Firefighting piping system is a critical utility network, safeguarding personnel, equipment, and facilities during both construction and operation. It consists of fire water mains, hydrants, hose reels, sprinkler lines, and monitors strategically located in high-risk areas such as turbine building, HRSG building, control electrical buildings, transformers, and fuel storage zones.
 - 2.3.4.2. Its primary function is to provide a dependable water supply at sufficient pressure and flow to support hydrants, sprinklers, deluge valves, and foam injection points, thereby preventing fire escalation and minimizing equipment damage.
 - 2.3.4.3. Installation begins at dedicated tanks, fire pumps, or municipal connections and is routed through trenches, and perimeters, with branch lines extending to critical areas where hydrants,

hose reels, and sprinkler systems are installed for rapid response.

2.3.4.4. Through early installation and proper operation, the system provides continuous fire protection, enabling safe construction progress, commissioning activities, and long-term plant safety.

2.3.5. Fuel supplying system

2.3.5.1. Fuel supply system delivers fuel from storage tanks or external pipelines to gas turbines, ensuring continuous and conditioned supply at the required pressure and flow. It consists of storage tanks, transfer pumps, filters, heaters, piping networks, preheating systems, and instrumentation, with safety devices such as isolation valves, check valves, relief valves, and emergency shutdown interlocks integrated to prevent hazards.

2.3.5.2. Fuel oil is pumped from storage through filtration to supply headers and routed to turbine fuel oil skids, where final conditioning and pressure control occur before reaching the nozzles.

2.3.5.3. Fuel gas piping is routed along pipe racks or trenches with proper supports and insulation downstream of preheaters to maintain temperature. Throughout the system, monitoring instruments, leak detection, and emergency shutoff valves are installed to ensure operational safety and reliability.

2.3.6. Demi. Water system

2.3.6.1. Demi. water piping system is a critical utility network that supplies high-purity water to process and auxiliary systems, including HRSGs, boilers, and steam turbines. Treated to remove dissolved salts and minerals, demi. water prevents corrosion, scaling, and fouling, thereby protecting sensitive equipment and sustaining plant reliability.

2.3.6.2. The system consists of storage tanks, transfer pumps, distribution headers, and piping designed to preserve water quality and ensure continuous delivery for process use, equipment flushing, and feedwater makeup. Installation begins at storage tanks with controlled quality and temperature, with water routed through pumps and main headers to designated process and auxiliary units.

2.3.6.3. Piping is laid along pipe racks or utility corridors, while branch lines incorporate valves, sampling points, and monitoring devices to ensure operational flexibility.

2.3.7. Potable Water system

2.3.7.1. Potable water piping system supplies safe, clean drinking water to administrative buildings, control rooms, restrooms, and first-aid stations. Designed to ensure hygienic access for personnel, it complies with health and safety regulations while preventing contamination through proper material selection, pressure control, and monitoring.

2.3.7.2. The network consists of storage tanks, pumps, piping, valves, and headers, with provisions for

emergency supply, backflow prevention, and pressure regulation to maintain reliability under varying demand conditions.

2.3.7.3. Water is sourced from storage tanks or municipal connections, treated and pressurized as necessary, then distributed through main headers along pipe racks, trenches, or corridors. Branch lines extend to points of use such as restrooms, eye-shower, and drinking fountains, equipped with valves, regulators, flow meters, and backflow prevention devices.

2.3.8. Service Water system

2.3.8.1. Service water piping system is a key utility network, supplying non-potable water for auxiliary, cooling, and general plant applications. It delivers water to equipment such as WT/WWT chemical material handling, ammonia tank shelter, cooling tower, chemical dosing shelter, HRSG blowdown tank and atmosphere flash tank quenching, and cleaning operations, ensuring adequate flow and reliable service.

2.3.8.2. The system consists of storage tanks, transfer pumps, distribution headers, and piping equipped with valves, strainers, and flow control devices to maintain operability and support maintenance. Water originates from storage tanks or raw water sources, then passes through pumps into main headers routed along pipe racks, trenches, or corridors to reach auxiliary equipment and washdown areas.

2.3.9. Instrument Air system

2.3.9.1. Instrument air system is a critical utility network, providing clean, dry, and oil-free compressed air to operate pneumatic instruments, control valves, actuators, and safety devices. It ensures accurate and reliable operation of turbine controls, HRSG systems, feedwater regulation, and auxiliary equipment, while supporting emergency shutdown systems, interlocks, and other safety-critical functions.

2.3.9.2. The network comprises air compressors, dryers, receivers, filters, piping, and distribution headers to maintain required pressure, flow, and air quality. Air is generated, filtered, dried, and stored in receivers before being distributed through main headers, with branch lines extending to end-use points equipped with isolation valves, pressure regulators, and check valves for safe and controlled delivery.

2.3.10. Service Air System

2.3.10.1. Service air piping system is a utility network that supplies compressed air for general plant operations and maintenance. Unlike instrument air, it supports pneumatic tools, valve actuation, cleaning, line blowdown, and other non-critical tasks.

2.3.10.2. The system comprises air compressors, receivers, piping, valves, and distribution headers to ensure reliable delivery throughout the plant. Compressed air is routed from compressors and

receivers through main headers, with branch lines extending to end-use points equipped with isolation valves, pressure regulators, and drain points for safe and controlled operation.

2.3.11. N2 Gas System

2.3.11.1. Nitrogen gas piping system is a utility network that supplies high-purity nitrogen to various systems and equipment in a combined cycle power plant. Nitrogen serves as an inert gas for pipeline purging, fuel storage tank blanketing, equipment protection, and operation of safety devices.

2.3.11.2. The system includes storage tanks or on-site generators, compressors, piping, valves, regulators, and distribution headers to ensure safe and reliable delivery. Nitrogen is routed from the source through main headers and branch lines, with isolation valves, check valves, flow meters, and safety relief devices installed at end-use points.

2.3.12. GT_ST_Misc. System

2.3.12.1. Fuel Oil & Fuel Gas Vent System

- Fuel oil and fuel gas vent system is a utility network that safely releases trapped gases or vapors from fuel supply lines and associated equipment in a combined cycle power plant. Its primary function is to prevent overpressure, remove air during startup, and discharge residual gases during shutdown, ensuring stable and safe fuel system operation.
- The system comprises vent piping, headers, valves, flame arrestors, and vent stacks, with branch lines routed from vent connections on fuel skids, filters, and high points to common headers leading to safe discharge locations.

2.3.12.2. GT & ST Lube Oil Vent System

- GT & ST lube oil vent system is a utility network that releases oil vapors and entrained gases from lubrication oil tanks, sumps, and bearing housings in a combined cycle power plant. Its primary function is to maintain stable pressure in lube oil reservoirs, prevent vapor accumulation, and ensure safe, continuous lubrication of gas turbines, steam turbines, and auxiliary equipment.
- The system comprises vent piping, headers, mist eliminators, flame arrestors, and vent stacks, with branch lines routed from lube oil tanks and bearing housings to common headers leading to the atmosphere or vapor treatment units.

2.3.12.3. GT Air Intake Anti-Icing System

- GT air intake anti-icing system is a specialized utility network designed to prevent ice formation on air filters, inlet louvers, and ducting in a combined cycle power plant. Its primary function is to maintain uninterrupted air flow, ensuring optimal combustion, turbine efficiency, and safe operation under cold weather conditions.

- The system comprises heaters, piping, valves, temperature sensors, and control devices, with heated air or hot water circulated through critical components and supported with brackets, anchors, and insulation to prevent ice accumulation and accommodate thermal expansion. Control and monitoring devices regulate heating based on ambient conditions, while corrosion-resistant materials, pressure relief valves, temperature alarms, and bypass arrangements ensure durability, safety, and long-term operational reliability.

2.3.12.4. Fuel Gas Pre heating System

- Fuel gas is delivered from the supply header to the turbine combustors through a conditioning and preheating system. The preheating system raises the temperature of natural gas using condensate extracted from the LP steam cycle, which transfers thermal energy to the fuel gas through a heat exchanger. This process improves combustion efficiency, ensures stable flame characteristics, and reduces thermal stress on turbine components.
- Fuel gas pre heating piping is routed from the extraction point to the preheater and back to the main condensate system, forming a reliable thermal circuit. The system is equipped with isolation valves, pressure control valves, bypass lines, and instrumentation for pressure, flow, and temperature monitoring, as well as safety devices such as leak detection and emergency shutdown valves.

2.3.12.5. Drain Piping System

- Drain piping system is a utility network designed to collect, convey, and remove rainwater, condensate, and incidental water from plant facilities and equipment. Its primary function is to prevent water accumulation and flooding, protecting equipment, electrical systems, and structural components while maintaining safe and efficient plant operation.
- The system consists of piping, floor drains, collection sumps, valves, and monitoring devices to ensure controlled and reliable drainage. Installation begins at floor drains, roof drains, and equipment sumps, where branch lines feed into main drain headers that route water to designated discharge points or stormwater systems.
- The piping is laid with appropriate slope along pipe racks, trenches, or enclosed conduits to facilitate gravity flow, and is supported with isolation valves, cleanouts, and access points to allow maintenance and inspection.

2.3.12.6. Waste Water Transfer System

- Wastewater transfer piping system is a specialized utility network that collects, conveys, and discharges chemical and oily effluents from plant operations, including lubrication oil drains, fuel treatment systems, chemical dosing units, and process cleaning activities.

Its primary function is to safely remove contaminated water, directing it to treatment units such as oil separators, neutralization tanks, or chemical treatment facilities to ensure environmental compliance and protect equipment.

- The network includes piping, pumps, valves, separators, and monitoring devices designed to maintain controlled and reliable transfer of effluents. Installation begins at designated collection points, where branch lines feed into main transfer headers equipped with separators or collection tanks, and pumps convey wastewater to treatment or discharge points.

2.4. Simple Cycle Operation (Gas Turbine)

- 2.4.1. Simple cycle operation refers to a system that uses only a single thermodynamic cycle, typically a gas turbine. In a simple cycle, the hot exhaust gas from the gas turbine is simply released into the atmosphere without being used to generate additional power. While simple cycle plants are less efficient, they are often used for "peaking power," meaning they can be started up quickly to meet sudden spikes in electricity demand.
- 2.4.2. For simple cycle operation, the main piping systems that need to be installed are those for fuel supply (Fuel Oil or Fuel Gas), the Closed Cooling Water (CCW) system for cooling major equipment such as the GT generator and its coolers, and utility systems for providing demi. water, service water, instrument air, GT air performance system and N2 Gas supplying system.
- 2.4.3. Simple Cycle operation centers around the Gas Turbine (GT). During this phase, the stability of the fuel supply and the integrity of the GT's own systems are paramount.

2.4.4. Gas Turbine and Ancillary Systems

- 2.4.4.1. Gas turbine system delivers efficient and reliable power as the first stage of the combined cycle process, while its hot exhaust gases are directed into the HRSG to produce steam.
- 2.4.4.2. Ancillary systems, including fuel supply, lubrication and hydraulic oil, cooling and sealing air, fire protection, and control and monitoring, support safe and stable turbine operation. These subsystems ensure the turbine operates within design parameters, minimizes downtime, and maintains high efficiency under varying load conditions. Air intake structures, equipped with filtration and silencing devices, provide clean air to the compressor.
- 2.4.4.3. The main gas turbine unit is mounted on a reinforced foundation within the GT building and directly coupled to the generator.
- 2.4.4.4. Fuel supply lines from external pipelines or storage tanks are routed to gas turbine skids, where valves and controls regulate pressure and flow.
- 2.4.4.5. Lubrication and hydraulic systems are installed on dedicated skids and piped to bearings, actuators, and other critical components.

- 2.4.4.6. Cooling and sealing air lines are distributed from compressor discharge or auxiliary blowers to turbine sections. Exhaust gas ducts, equipped with insulation and expansion joints, convey hot gases from the turbine to the HRSG while accommodating thermal stresses.
- 2.4.4.7. Additional ancillary systems, such as fire detection and suppression, enhance operational safety, ensuring reliable power generation and overall combined cycle efficiency.

2.4.5. Fuel Supply Systems (Fuel Oil or Fuel Gas)

- 2.4.5.1. Fuel supply system is the critical lifeline for the gas turbine, where any leak could result in catastrophic incidents, making precise construction and installation essential. In combined cycle power plants, natural gas is typically used for turbine startup, but diesel or heavy oil may be employed depending on supply conditions; where natural gas infrastructure is not yet available, ultra-low-sulfur diesel (ULSD) may serve as the primary fuel until the gas supply is established.
- 2.4.5.2. All fuel oil welds and connections undergo 100% NDE for butt joints to detect defects and ensure safety. Fuel oil is received into buffer storage tanks, then routed to a treatment shelter for purification before being pressurized, filtered, and delivered to the fuel oil skid for combustion preparation.
- 2.4.5.3. Piping and associated facilities are designed for fuel compatibility, incorporating proper drainage and spill containment to safely handle oily water and prevent environmental contamination.
- 2.4.5.4. For fuel gas, high-pressure lines require materials and thicknesses meeting design specifications, with full penetration welds verified by hydro testing and vent piping routed to elevated safe locations.
- 2.4.5.5. Natural gas is supplied from the OSBL to a governor station for pressure regulation, passes through performance heaters to optimize temperature, and flows through a final filter before entering the fuel gas skid and turbine.
- 2.4.5.6. High-pressure gas lines follow stringent quality control standards, including welding procedures, NDE, and safe vent installation to prevent leaks and ensure operational safety.
- 2.4.5.7. The system integrates with the GT air performance system, using exhaust heat to warm fuel gas between the HRSG economizer and the fuel gas performance heater. This configuration ensures reliable, safe, and efficient fuel delivery, supporting continuous and stable gas turbine operation.

2.4.6. Lube Oil & Cooling System

- 2.4.6.1. GT Lube Oil and Cooling System piping is a critical network that delivers lubricating oil to turbine bearings, gearboxes, and auxiliary rotating equipment while providing cooling to

maintain safe operating temperatures. Its primary function is to supply clean, pressurized oil continuously, lubricating moving parts, removing frictional heat, and supporting auxiliary systems such as filtration, deaeration, and temperature control to maintain oil quality.

- 2.4.6.2. The system ensures operational reliability by protecting critical equipment during startup, operation, and shutdown. Oil is stored in the lube oil reservoir and pumped through main and auxiliary pumps to oil coolers, filters, and control valves before being routed to bearings, gearboxes, and auxiliary equipment, then returned to the reservoir for recirculation.
- 2.4.6.3. Piping is installed near the turbine skid, in trenches, on pipe racks, or on turbine-mounted supports to facilitate maintenance and inspection, with proper slope in return lines to ensure smooth drainage. Critical sections are equipped with temperature and pressure sensors, isolation valves, and bypass lines to secure uninterrupted supply and safe operation. Hangers and supports are provided to prevent vibration-induced fatigue, while the lube oil tank vent is routed to a safe location, typically the GT building roof.
- 2.4.6.4. CCW system supplies cooling water from the cooling tower to heat exchangers, including the lube oil cooler, maintaining oil within safe temperature limits. CCW lines are sized to provide sufficient flow for effective heat removal.

2.4.7. Instrument Air (IA) & Nitrogen (N2) Gas Systems

- 2.4.7.1. Instrument air is used for actuating pneumatic valves, operating control devices, and supporting instrumentation, while nitrogen provides an inert atmosphere for purging fuel lines, blanketing storage tanks, and maintaining safe pressure in critical areas. Together, these gases ensure reliable turbine operation, precise control, and operational safety. The instrument air system supplies moisture-free, oil-free air to actuators, control valves, and instruments, enabling accurate turbine control, fuel regulation, and protection functions.
- 2.4.7.2. Instrument air piping is typically installed along pipe racks or in trenches, terminating at local distribution manifolds with branch connections to valves, actuators, and instrumentation panels.
- 2.4.7.3. Nitrogen piping delivers inert gas to fuel skids, purge points, and sensitive process areas, reducing the risk of combustion or explosion. Installation begins at air compressors and dryers for instrument air, and at nitrogen storage vessels or bottle rack for nitrogen, from which gases are routed through headers and branch lines to the turbine skid and auxiliary systems.
- 2.4.7.4. Nitrogen piping is equipped with regulators, isolation valves, and check valves to control flow and pressure, while both systems include monitoring instruments, pressure relief devices, and leak detection to ensure safe operation.

2.4.8. GT Air Intake System

- 2.4.8.1. GT Air Intake System is designed to supply a continuous flow of clean, debris-free air to the turbine compressor while preventing ice formation under cold or humid conditions. Air passes through inlet filters, silencers, and ducts to ensure that dust, particulates, and contaminants are removed before reaching the compressor.
- 2.4.8.2. The anti-icing subsystem delivers heated air or steam, typically extracted from the turbine compressor or an auxiliary source, to inlet guide vanes and plenum surfaces to prevent ice accumulation and maintain airflow.
- 2.4.8.3. The piping network includes main headers and branch lines connected to distribution manifolds, ensuring uniform heating across all critical surfaces. Control valves, regulators, and temperature sensors manage the flow and temperature of the heating medium, while isolation valves allow maintenance without interrupting turbine operation. Drain lines and safety relief devices are integrated to remove condensate and protect the system from overpressure.
- 2.4.8.4. The air intake and anti-icing piping are routed along the intake duct using structural brackets and hangers, with thermal insulation applied to maintain efficiency and minimize heat loss.
- 2.4.8.5. Installation positions the intake structure at an elevated level to reduce debris ingestion, and the compact layout of the piping ensures minimal pressure drop.

2.4.9. GT Exhaust System

- 2.4.9.1. GT Exhaust System piping is designed to safely collect, channel, and discharge high-temperature combustion gases generated during turbine operation. It consists of the exhaust diffuser, main ducting, expansion joints, support structures, and, when applicable, bypass stacks or diverter dampers, all fabricated from heat-resistant materials to withstand continuous thermal cycling.
- 2.4.9.2. The primary function is to guide exhaust gases either to the HRSG in combined cycle operation or to the atmosphere via a bypass stack in simple cycle mode, ensuring controlled flow with minimal pressure loss. Exhaust ducting is routed from the turbine diffuser outlet through horizontal or slightly inclined paths toward the HRSG inlet, using transition pieces to match geometries and maintain flow efficiency.
- 2.4.9.3. Expansion joints are strategically placed to accommodate thermal growth and vibration, while support frames and spring hangers allow controlled movement without compromising stability.
- 2.4.9.4. Diverter dampers, when included, redirect flow between the HRSG and bypass stack depending on operational mode. Instrumentation lines, such as thermocouples and pressure transmitters, are installed to monitor exhaust conditions, and access platforms and manholes provide inspection and maintenance points.

- 2.4.9.5. High-temperature insulation is applied along the duct to reduce heat loss, protect adjacent structures, and improve overall system efficiency.
- 2.4.9.6. The exhaust system incorporates noise attenuation, emission control, and thermal management features to comply with safety and environmental standards.

2.5. Combined Cycle Operation (HRSG & ST Operation)

- 2.5.1. Combined cycle is an integrated power generation process that couples a gas turbine and a steam turbine to maximize efficiency and minimize fuel consumption. The gas turbine generates electricity through fuel combustion while simultaneously producing high-temperature exhaust gas. Instead of being released into the atmosphere, this exhaust is directed to a HRSG, where it is converted into steam to drive a steam turbine for additional power generation.
- 2.5.2. By reusing waste heat, the combined cycle achieves significantly higher efficiency than simple cycle plants, often exceeding 55–60%. This improvement not only reduces fuel consumption but also lowers greenhouse gas emissions, making the system both cost-effective and environmentally sustainable.
- 2.5.3. The combined cycle arrangement also enhances operational flexibility, allowing stable performance under varying grid demands. Installation begins with the gas turbine, mounted on a reinforced foundation and coupled directly to the generator.
- 2.5.4. Exhaust gases flow through insulated ductwork into the HRSG, which is typically positioned adjacent to the turbine to minimize heat loss and accommodate expansion.
- 2.5.5. Steam produced in the HRSG is routed via main steam lines to the steam turbine, aligned either in a single-shaft or multi-shaft configuration depending on plant design.
- 2.5.6. Supporting systems—including condensate, feedwater, cooling water, and auxiliary fuel and electrical networks—are routed through pipe racks and utility corridors to ensure seamless integration, efficient energy transfer, and maintainable plant operation.

2.5.7. HRSG System

- 2.5.7.1. HRSG is a core component that links the gas turbine and steam turbine in a combined cycle power plant. It recovers the high-temperature exhaust gases from the gas turbine to generate steam instead of releasing the heat to the atmosphere. By capturing this energy, the HRSG significantly increases overall thermal efficiency compared to simple cycle operation.
- 2.5.7.2. The system produces steam at multiple pressure levels—high, intermediate, and low—which is supplied to the steam turbine for additional electricity generation. It also provides reheated steam for improved efficiency, auxiliary steam for startup and plant operations, and blowdown facilities for maintaining water quality.
- 2.5.7.3. The installation route begins at the gas turbine exhaust outlet, where insulated ducts transfer

flue gases into the HRSG inlet. Inside the HRSG, gases pass through economizer, evaporator, and superheater sections, while the generated steam is collected in drums and routed to the corresponding turbine inlets. Reheat steam piping connects the reheater section to the steam turbine between HP and IP stages, while condensate and feedwater circuits are integrated through economizers and evaporators.

- 2.5.7.4. Piping networks are arranged on pipe racks with expansion joints, insulation, and access for maintenance, supported by safety devices such as valves, vents, drains, and instrumentation.

2.5.8. HRSG Steam System (HP, HRH, CRH, LP Steam)

- 2.5.8.1. HRSG steam system forms the essential link between the gas turbine exhaust and the steam turbine in a combined cycle power plant. It converts high-temperature exhaust gases into steam at multiple pressure levels—high, intermediate, and low—thereby transforming waste heat into useful energy.
- 2.5.8.2. HP steam delivers the highest energy steam generated in the HRSG, where exhaust gas from gas turbine heats the steam in the superheater to the required temperature and pressure. From the HP steam drum, the steam flows through the superheater, passes the stop and control valves, and enters the HP turbine casing to provide the main driving force for the steam cycle. Piping is engineered to withstand extreme thermal and pressure conditions, incorporating expansion loops or compensators to manage thermal growth.
- 2.5.8.3. IP steam system generates medium-pressure steam to optimize turbine performance and balance the cycle, supporting both additional power generation and part-load flexibility. Steam is collected from the IP drum, heated in the superheater sections, and routed through control and stop valves to the IP turbine inlet. Depending on plant configuration, the IP system may also integrate with the reheat system to further enhance efficiency.
- 2.5.8.4. Reheat steam system enhances turbine efficiency by returning the HP turbine exhaust steam (cold reheat, CRH) to the HRSG reheater, where it is reheated using gas turbine exhaust, and the reheated steam (hot reheat, HRH) is then directed to the IP turbine for further expansion. The reheat piping runs from the HP turbine exhaust through the HRSG reheater to the IP turbine inlet, equipped with stop and intercept valves for flow control and safe shutdown.
- 2.5.8.5. LP steam system utilizes residual heat from the gas turbine exhaust to generate low-pressure steam in the HRSG evaporator and superheater sections, which is then supplied to the LP turbine for expansion. Its design maximizes energy recovery by converting remaining thermal energy into useful power. Due to the higher volumetric flow, LP steam lines are larger in diameter and routed from the LP steam drum through the superheater to the turbine inlet, equipped with isolation and control valves.

- 2.5.8.6. Auxiliary steam headers branch off from the main lines to supply sealing steam to turbine glands, provide steam for start-up operations, or support auxiliary heating systems.
- 2.5.8.7. Steam generated in HRSG drums is routed through insulated piping networks equipped with stop valves, control valves, drains, vents, and monitoring instruments.
- 2.5.8.8. Piping is typically routed on elevated racks with sufficient space for thermal expansion and safe access for inspection and maintenance. Expansion joints, spring hangers, and constant load hangers are applied to accommodate thermal movement, while piping materials are designed to withstand extreme temperatures and pressures.
- 2.5.8.9. Weld integrity is verified by NDE, such as RT and Ultrasonic Testing (PAUT).
- 2.5.8.10. Prior to commissioning, hydro testing confirms system strength and reliability, and chemical dosing lines are installed to maintain water chemistry, preventing corrosion and scaling of boiler components.

2.5.9. Feed Water System

- 2.5.9.1. Feedwater system is a vital component of the steam-water cycle in a combined cycle power plant, ensuring a continuous supply of high-purity water to the HRSG. It sustains steam generation by delivering properly conditioned water under all load conditions, making it indispensable for reliable plant operation.
- 2.5.9.2. The system recovers condensate from the condenser, treats it, and pressurizes it before returning it to the HRSG.
- 2.5.9.3. During this process, the condensate is heated in the condensate preheater (CPH) or the LP economizer section of the HRSG, where it absorbs heat from the gas turbine exhaust to enhance cycle efficiency, instead of from steam extracted from the turbine as in a conventional steam plant. The water then enters the deaerator (a separate unit or the deaerating section integral to the HRSG LP drum, depending on the HRSG design), which eliminates dissolved gases such as oxygen and carbon dioxide to prevent corrosion. The boiler feed pumps (BFP) subsequently raise the water pressure to the required level for delivery into the HRSG HP and IP economizers, where the feedwater is further heated by the exhaust gas before entering the drums and evaporator sections.
- 2.5.9.4. The installation route begins at the condenser hot well, where the condensate extraction pumps transfer water through the condensate polishing units and the condensate preheater to the deaerator or the LP drum. From the deaerator (or the LP drum where the deaerator is integral to the HRSG), which is positioned at an elevated level to ensure sufficient NPSH for the boiler feed pumps, the water is pumped through the feedwater control valves to the HRSG HP and IP economizer inlets.

- 2.5.9.5. Piping is routed along pipe racks and turbine deck structures, with thermal expansion provisions, insulation, and maintenance access incorporated into the layout.
- 2.5.9.6. Control valves, flow meters, and safety devices are strategically installed to regulate pressure, safeguard water chemistry, and maintain safe operation.

2.5.10. Condensate Water System

- 2.5.10.1. Condensate system is a critical component of the thermal cycle in a combined cycle power plant, responsible for collecting and transferring condensate from the condenser back to the feedwater system. Its primary function is to maintain continuous recovery and circulation of high-purity water, ensuring the efficiency and reliability of the steam cycle.
- 2.5.10.2. By returning condensed steam from the low-pressure turbine to the cycle, the system minimizes treated water loss and reduces the requirement for make-up demineralized water.
- 2.5.10.3. This process also preserves stable water chemistry, preventing corrosion and scaling in the HRSG, turbines, and other critical equipment.
- 2.5.10.4. The installation route typically begins at the condenser, where exhaust steam from the LP turbine is condensed into liquid form.
- 2.5.10.5. Condensate pumps then transfer the water through the polishing units and the condensate preheater (CPH) to the deaerator (where provided) or the HRSG LP drum, before delivery to the feedwater system.
- 2.5.10.6. Along the route, control valves, instrumentation, and protection devices regulate flow, pressure, and water quality to ensure safe and efficient operation.
- 2.5.10.7. Piping network is generally routed below the turbine floor and along designated pipe racks, optimizing pressure drop and allowing ease of access for maintenance.
- 2.5.10.8. Expansion joints and supports accommodate thermal growth, vibration, and system dynamics.

2.5.11. Sampling System

- 2.5.11.1. Sampling system is a critical auxiliary network designed to continuously monitor and control water and steam quality within the HRSG. Proper water chemistry is essential in high-pressure steam and feedwater circuits to prevent corrosion, scaling, and deposition that could damage turbine blades, HRSG tubes, and other pressure components.
- 2.5.11.2. The system enables operators to obtain accurate, real-time data for effective plant operation and equipment protection.
- 2.5.11.3. Its primary function is to extract representative samples from key process points, including the condensate line, feedwater line, economizer inlet, steam drums, superheater outlet, and reheater outlet. Samples are cooled, conditioned, and analyzed for parameters such as pH,

conductivity, dissolved oxygen, silica, phosphate, sodium, and chloride, allowing timely corrective actions like chemical dosing or blowdown.

2.5.11.4. By maintaining continuous monitoring, the system preserves equipment integrity, extends component life, and ensures compliance with plant chemistry specifications.

2.5.11.5. The installation route begins at multiple sampling points on steam and water lines, where small-bore stainless steel tubing is routed to a central sampling panel or skid.

2.5.11.6. High-temperature and high-pressure samples pass through coolers, pressure reducers, and flow control devices before reaching analyzers, display units, or sampling rack collection points.

2.5.11.7. Sampling lines are typically installed along pipe racks and directed to a dedicated sampling room with controlled environmental conditions for precise measurement.

2.5.12. Chemical Dosing System

2.5.12.1. Chemical dosing system is a critical auxiliary network in a combined cycle power plant that ensures proper control of water and steam chemistry within the HRSG, condenser, and feedwater circuits.

2.5.12.2. Operating under high temperature and pressure, these circuits require precise chemical treatment to prevent corrosion, scaling, and deposition, which could compromise equipment performance and plant reliability.

2.5.12.3. Its primary function is to inject controlled amounts of conditioning agents, such as ammonia or amines for pH control, hydrazine or oxygen scavengers, and phosphate, into the water-steam cycle at designated points including condensate lines, feedwater lines, and HRSG drums.

2.5.12.4. This process neutralizes impurities, minimizes corrosion, and prevents scaling, thereby extending the service life of HRSGs, turbines, and associated piping. The system also maintains plant efficiency and ensures compliance with regulatory water and steam quality standards.

2.5.12.5. Installation route begins at chemical storage tanks located in a dedicated handling area, from which chemicals are transferred to day tanks or dosing tanks for concentration adjustment.

2.5.12.6. Metering pumps then inject chemicals into small-bore lines directed to the target process points, equipped with quills or nozzles to ensure uniform mixing.

2.5.12.7. Piping and tubing are routed along pipe racks or trenches with proper containment to prevent leaks or environmental hazards. Control panels, flow meters, and monitoring instruments regulate dosing rates and provide feedback to operators.

2.6. Steam Turbine Generator (STG) & Condenser System

2.6.1. STG Steam System

2.6.1.1. Steam system within the STG building comprises main steam lines, auxiliary steam lines, and

associated equipment required for safe and controlled steam distribution. This system serves as the central hub where high-pressure, intermediate-pressure, and low-pressure steam generated by the HRSG is routed, conditioned, and delivered to the steam turbine.

- 2.6.1.2. Critical valves, headers, and safety devices are installed to manage pressure, flow, and temperature before steam enters the turbine.
- 2.6.1.3. Its primary function is to transfer steam from the HRSG to the turbine efficiently while providing pressure reduction, flow control, and protection through stop valves, control valves, and safety relief devices.
- 2.6.1.4. The system also supports auxiliary functions including gland sealing, turbine warm-up, and bypass during start-up and shutdown, ensuring stable and reliable plant operation.
- 2.6.1.5. The installation route begins with the main steam lines entering from the HRSG outlet headers and is routed to high-pressure stop and control valves before connecting to the turbine casing.
- 2.6.1.6. Intermediate- and low-pressure steam lines follow designated paths to their respective turbine stages, passing through reheaters and control points as needed, while auxiliary lines supply gland seals, auxiliary heaters, and the deaerator (where provided).
- 2.6.1.7. Piping is supported on embedded supports, hangers, and spring supports to accommodate thermal expansion, and all lines are insulated for efficiency and personnel safety.
- 2.6.1.8. Safety, stop, and control valves are strategically positioned to regulate flow and pressure, and piping is routed above ground on racks or dedicated supports to minimize heat loss.
- 2.6.1.9. Rigorous quality control ensures precise alignment and integrity of all connections, preventing stress on turbine nozzles and guaranteeing reliable, long-term steam delivery.

2.6.2. Bypass Steam System

- 2.6.2.1. Bypass steam system is an auxiliary network designed to divert steam from the HRSG directly to the condenser or auxiliary headers without passing through the steam turbine. It provides a controlled pathway for steam during abnormal or startup conditions, protecting turbine equipment and maintaining operational flexibility.
- 2.6.2.2. Its primary function is to relieve the turbine from receiving steam when full operation is not desired, such as during start-up, shutdown, or sudden load changes.
- 2.6.2.3. By bypassing the turbine, the system prevents thermal stress, reduces turbine wear, and allows the HRSG to continue operating without interruption. It also supports turbine testing, controlled warm-up procedures, and emergency pressure relief, ensuring that steam pressure in the HRSG and piping remains within safe limits.
- 2.6.2.4. The installation route begins at key points on the main steam lines or turbine stop valves,

where bypass piping is connected through dedicated stop and control valves.

- 2.6.2.5. From these points, HP bypass steam is generally routed to the cold reheat (CRH) line, while HRH and LP bypass steam is routed to the condenser, with additional branches to auxiliary headers or the deaerator (where provided) as needed to handle high-, intermediate-, or low-pressure steam.
- 2.6.2.6. The piping network includes pressure relief devices, drains, vents, and instrumentation to monitor flow and pressure, ensuring safe operation. All lines are supported on pipe racks or within the steam building, with proper insulation and allowances for thermal expansion.

2.6.3. Steam Drain System

- 2.6.3.1. Steam drain system is an essential auxiliary network that collects and removes condensate formed in steam lines, turbines, and auxiliary equipment of a combined cycle power plant.
- 2.6.3.2. Condensate forms as steam flows through superheaters, reheaters, headers, and piping due to temperature drops or pressure losses.
- 2.6.3.3. The primary function of the system is to maintain steam quality and protect equipment by continuously draining this condensate from main steam lines, turbine inlets and exhausts, reheaters, superheaters, steam tracing lines, and auxiliary systems.
- 2.6.3.4. Drained condensate is directed to condensate return lines or the feedwater system, allowing water to be reused and preventing disruptions caused by water slug formation or pressure surges.
- 2.6.3.5. Installation route begins at low points, equipment inlets, and pipe bends where condensate tends to accumulate, with small-bore drain lines connected to condensate headers or traps. Steam traps, either mechanical or thermodynamic, separate condensate from live steam, while vent lines release non-condensable gases during startup or shutdown.
- 2.6.3.6. Condensate flows by gravity or low-pressure pumps to collection tanks or the feedwater system, with piping insulated where necessary. All lines are supported on pipe racks or hangers and installed with proper slope to ensure complete drainage and prevent water hammer. Venting and instrumentation are strategically installed to monitor flow and ensure safe operation.

2.6.4. Circulation Water (CW) System

- 2.6.4.1. Condenser cooling water system relies on large-bore pipes designed to circulate cooling water efficiently. These pipes are typically constructed from GRP in underground to rubber lining carbon steel in above ground to withstand mechanical stress and thermal effects.
- 2.6.4.2. Due to their substantial weight, robust support structures and hangers are installed to prevent sagging and maintain system stability. The system receives a continuous supply of cold water

from the cooling tower, delivered by the main cooling water pumps.

- 2.6.4.3. Proper flow distribution ensures effective heat removal from the condenser and maintains turbine efficiency. Expansion joints are incorporated to accommodate thermal expansion and contraction of the piping during operation. Supports and anchors are strategically positioned to manage both static and dynamic loads. Piping alignment is critical to prevent undue stress at connection points and ensure long-term reliability.

3. CCPP Key Milestone

3.1. Piping Work based on Key Milestones

3.1.1. Establishing Foundational Utility Systems

- 3.1.1.1. Piping systems required at Initial infrastructure construction stage primarily include utility systems, such as temporary fire water lines, service water lines, demineralized water lines, instrument and service air, and fuel supply connections to support construction equipment, testing, and early commissioning activities. These systems enable safe hot work, hydrostatic testing, and equipment pre-commissioning.

3.1.2. Power Receiving

- 3.1.2.1. Power receiving involves establishing electrical connections between the plant and the utility grid or temporary construction power supply. This ensures that the plant has a stable power source for lighting, tools, auxiliary equipment, and initial testing activities.
- 3.1.2.2. Piping systems associated with this phase include firefighting piping to protect electrical equipment and transformers during energization and testing, and potable water lines for the restrooms in WT/WWT and the Control Electrical Building (CEB) to support on-site personnel.

3.1.3. Demi. Water Production

- 3.1.3.1. During the Demi. water production phase, the plant establishes a high-purity water system that supplies treated water for the feedwater system, HRSG make-up, and auxiliary equipment flushing.
- 3.1.3.2. The piping systems required include the demineralized water piping network, transfer pumps, storage tanks, and distribution headers. This system ensures continuous delivery of high-purity water to the HRSG and auxiliary systems, protecting sensitive equipment from scaling, corrosion, and contamination during start-up and testing.

3.1.4. Cooling Water System Load Test

- 3.1.4.1. Before gas turbine start-up and steam cycle operation, the cooling water systems, including the closed cooling water (CCW), main cooling water (MCW), and auxiliary cooling water (ACW) systems, are flushed, hydro-tested, and load tested together with the cooling tower. The

piping systems required include the cooling tower basin and pump house piping, the condenser water box connections, and the heat exchanger supply and return lines, which must be fully installed and supported, with expansion joint tie-rods secured, beforehand.

3.1.5. HRSG Hydro Test, Condensate and Feedwater Pump Load Test, and Chemical Cleaning

3.1.5.1. The HRSG pressure parts and the connected steam, feedwater, and condensate piping are hydro-tested with demineralized water. The condensate extraction pumps (CEP), condensate pre-heater (CPH), and boiler feed pumps (BFP) are then flushed and load tested, and the HRSG steam, feedwater, and condensate systems are chemically cleaned to remove mill scale, oil, and welding debris. Temporary piping, blinds, vents, and drains are required, and the permanent piping and supports of these systems must be completed beforehand.

3.1.6. GT Lube Oil Flushing and Functional Test

3.1.6.1. Before initial firing, the GT lube oil system is flushed and the GT auxiliary systems are functionally tested. The piping systems required include the lube oil piping and temporary flushing loops, closed cooling water, instrument air, fuel supply, and drain lines. The fuel supply piping must be hydro-tested, cleaned, and purged with nitrogen, and the CCW and IA lines to the generator, lube oil cooler, and other auxiliary coolers must be completed before the turbine is started.

3.1.7. GT Initial Firing

3.1.7.1. GT initial firing is the first ignition of fuel in the turbine combustors to confirm operational readiness. This phase requires the fuel supply piping system, including fuel storage tanks, transfer pumps, filtration and heating systems, and piping to the gas turbine skid. Auxiliary utility piping such as instrument air, service air, and cooling water lines are also required to operate control systems, valve actuators, and cooling circuits during turbine startup. Safety piping systems, including firefighting lines, are essential to ensure secure operation during initial firing.

3.1.8. GT Synchronization

3.1.8.1. GT synchronization is the process of connecting the gas turbine-generator to the electrical grid in simple-cycle operation after initial firing. The piping systems involved primarily include fuel supply, lube oil, cooling water, and instrument air lines, together with drain, vent, and firefighting water lines, to maintain stable turbine and generator operation during load-up.

3.1.9. Steam Blowing

3.1.9.1. Steam blowing is the process of cleaning the HRSG and the steam piping upstream of the steam turbine to remove debris, mill scale, and other contaminants. During this stage, the steam system piping is pressurized and operated, including main steam lines, drain lines, and

bypass steam lines.

- 3.1.9.2. The demi. water system supplies water for generating steam in the HRSG. Safety valves, drain headers, and blowdown lines are operated to safely discharge debris-laden steam to designated areas, ensuring that the steam path is clean before turbine operation.

3.1.10.ST Lube Oil Flushing

- 3.1.10.1. Before the steam turbine is rolled, the lube oil system is flushed to remove construction debris and protect the turbine bearings. The piping systems required include the lube oil piping from the reservoir to the bearings, the lube oil cooler cooling water lines, temporary flushing lines and bypasses, and the associated instrumentation and drain lines, all of which must be completed and hydro-tested beforehand.

3.1.11.ST Steam Admission

- 3.1.11.1. Steam turbine steam admission phase involves gradually admitting steam from the HRSG into the steam turbine to begin controlled operation. The piping systems required include main steam lines, intermediate and low-pressure steam lines, control valves, bypass lines, steam drain lines, and auxiliary steam lines for turbine warming and seal supply.
- 3.1.11.2. Proper operation of the piping ensures that steam reaches the turbine at controlled pressure and temperature, minimizing thermal stresses and protecting the turbine from damage.

3.1.12.ST Synchronization

- 3.1.12.1. ST synchronization is the process of connecting the steam turbine-generator set to the electrical grid in combined-cycle operation after steam admission. The piping systems involved in this phase primarily include condensate return lines, feedwater lines, auxiliary steam lines, and cooling water lines to maintain turbine and generator operation at safe temperatures and pressures during synchronization. Instrumentation and control air piping are also critical to operate valves and sensors accurately during grid connection.

3.1.13.Performance & Reliability Tests

- 3.1.13.1. During performance and reliability testing, the plant is operated at various loads to verify that all systems function correctly and efficiently under design conditions. Piping systems active in this phase include main steam lines, feedwater lines, circulation water lines, demineralized water lines, fuel supply lines, chemical dosing lines, and all utility systems. These systems are monitored for pressure, flow, temperature, and integrity to ensure compliance with performance specifications and reliability criteria.

3.1.14.Commercial Operation Date (COD)

- 3.1.14.1. Commercial Operation Date marks the official handover of the plant to operational status for continuous power generation.

3.1.14.2. All piping systems, including main process lines (steam, feedwater, fuel), circulation and cooling water systems, demineralized water systems, instrument and service air, service water, firefighting, and utility systems, must be fully commissioned and operational.

3.1.14.3. The plant is now capable of sustained operation, meeting design output, efficiency, and environmental requirements.

3.2. Detailed Guide to Commissioning with Piping Schedule Integration

3.2.1. Each milestone is directly linked to the completion of specific piping systems. Piping construction schedule must be managed to meet these critical preconditions for commissioning activities.

3.2.2. Establishing Foundational Utility Systems

3.2.2.1. Initial water and power supply is about establishing the fundamental infrastructure for plant safety and operation.

3.2.3. Fire Fighting and Potable Water Supply

3.2.3.1. Firefighting system includes underground and aboveground piping, hydrants, hose reels, and sprinkler networks, designed to deliver water promptly to protect personnel, equipment, and facilities during fire emergencies.

3.2.3.2. Firefighting piping typically originates from fire water storage tanks or pump houses, with large-diameter underground lines distributing water across the plant and rising aboveground at hydrants, hose reels, and sprinkler headers. Fire pumps, jockey pumps, and pressure maintenance systems maintain adequate flow and pressure throughout the network. The fire water storage tank or reservoir is typically located at the periphery of the plant, and large-diameter underground or aboveground mains are routed along the plant's main utility corridors, with branch pipelines extending from the mains to cover all process areas, turbine buildings, HRSG structures, control buildings, fuel storage facilities, and administrative zones.

3.2.3.3. Hydrants are strategically located along access roads and around high-risk areas to provide effective coverage. Sprinkler and hose reel lines are routed within enclosed buildings and electrical rooms to protect sensitive equipment.

3.2.3.4. Normally, piping work such as HDPE piping installation in the underground and fire water supplying piping from outside resource to fire water storage tank and fire firefighting pump house and fire water distribution lines to each building sprinkler and hydrant would be proceed by piping team.

3.2.3.5. Other piping work such as hydrants, hose reels, and sprinkler headers, fire pumps, jockey pumps, and pressure maintenance systems are installed by specific firefighting construction vendor.

- 3.2.3.6. Potable water piping provides treated water for drinking, sanitation, and other domestic and operational needs, ensuring compliance with health and safety standards.
- 3.2.3.7. Potable water piping begins at treatment units or municipal connections, routing underground to main facilities and branching inside buildings to supply restrooms, eye-wash stations, and welfare areas.
- 3.2.3.8. Materials for firefighting piping include carbon steel, hot-dip galvanized with internal and external coatings where specified to prevent corrosion, or HDPE for underground runs, while potable water lines use stainless steel or food-grade polyethylene/PVC and HDPE for underground as appropriate.
- 3.2.3.9. Both systems are equipped with valves, fittings, strainers, and gauges, while the firefighting system additionally integrates hydrants, sprinklers, and other suppression equipment. The firefighting system also includes main supply lines, branch distribution lines, hydrant connections, sprinkler and hose reel piping, pump discharge piping from the fire water pump house, jockey pump lines, test lines, non-return valves, pressure regulating devices, and instrumentation lines, which are installed to maintain operational readiness.
- 3.2.3.10. Installation and pressure testing of both systems must be completed prior to power commissioning to ensure operational readiness and compliance with safety regulations. In particular, the firefighting water system is one of the earliest utility installations, as it secures a reliable and immediate water supply for fire suppression and is essential for both worker safety and compliance with fire protection standards. It must be installed and commissioned before power receiving so that fire protection is available during construction and pre-commissioning, and it is a prerequisite for power receiving and subsequent commissioning activities.
- 3.2.3.11. Potable Water piping to areas like WT/WWT and Control Electrical Building (CEB) bathroom must be completed prior to power receiving to support on-site personnel.

3.2.4. Fresh Water Supply

- 3.2.4.1. Fresh water supply piping system in a combined cycle power plant is designed to provide raw water for various plant operations and utility services. This system typically delivers water prior to any treatment processes, serving as the main source for producing Demi. water, service water.
- 3.2.4.2. Fresh water supply piping is to transfer water from an external source, such as a municipal connection, river intake, or dedicated raw water reservoir, to the plant's water treatment facilities.
- 3.2.4.3. This system ensures an uninterrupted water supply for essential processes, including demi.

water production for boilers and turbines, service water for cooling and washing.

- 3.2.4.4. The installation route of fresh (reuse) water supply piping begins at the plant's intake point, which may be connected to municipal pipelines or a raw water pumping station. From this intake underground pipelines are routed toward the fresh water storage tanks or water treatment units. From the storage tanks, distribution branches are installed to deliver water to different plant facilities such as the demineralization unit, service water supply.
- 3.2.4.5. Piping is laid along designated utility corridors and pipe racks, ensuring accessibility for maintenance and inspection.
- 3.2.4.6. Fresh water supply system generally uses corrosion-resistant materials, such as lined carbon steel or stainless steel depending on water quality, to prevent corrosion and scaling. In certain cases, HDPE pipes are employed for underground installation due to their durability and resistance to soil conditions.
- 3.2.4.7. The system incorporates isolation valves, strainers, and non-return valves to ensure proper flow control and contamination prevention. Connection points are provided for routing to auxiliary systems, and metering devices are installed to monitor the quantity and quality of water supplied to each facility.
- 3.2.4.8. Fresh Water is the raw water source for demi. water production. Therefore, the piping to the water treatment building must be completed on a high-priority basis.

3.2.5. Power Receiving

- 3.2.5.1. During power receiving, critical piping systems must be installed and fully operational to support the transition from construction to commissioning. This stage allows the plant to receive electrical power from the grid or external sources, energizing auxiliary systems, pumps, and key equipment.
- 3.2.5.2. Piping works focus on completing essential utility networks and ensuring the readiness of process lines required for gas turbine firing and plant start-up. The primary function of these piping systems is to provide reliable supply and circulation of utilities such as fire water, potable water, service water, demineralized water, instrument air, and fuel gas or oil.
- 3.2.5.3. These networks support auxiliary equipment, lubrication systems, cooling circuits, and safety systems to ensure proper operation once electrical power is introduced. They also maintain safe working conditions during energized commissioning by securing fire protection, flushing utilities, and maintaining cooling and lubrication readiness.
- 3.2.5.4. Piping installation begins with main headers and branch lines that interconnect utility systems with the respective equipment. Routes are designed to be accessible and supported on pipe racks or installed underground according to plant layout requirements. Proper coordination

with instrumentation, valves, and control systems is implemented to allow smooth operation and monitoring.

3.2.5.5. The piping systems required during the power receiving stage include the following:

- Fire Fighting Water System: Distribution mains, hydrant connections, sprinkler lines, and hose reels.
- Potable and Service Water Systems: Piping from storage tanks to domestic and utility service points.
- Demi. Water System: Supply lines from the water treatment plant to demi. water tanks and auxiliary equipment.
- Instrument Air and Service Air Systems: Distribution headers, branch lines, and connections to pneumatic devices.
- Fuel Supply System: High-pressure gas or liquid fuel oil supply lines, treatment lines, and connections to turbine fuel skids.
- Auxiliary Cooling and Lube Oil Support Lines: Piping that ensures proper lubrication and cooling of rotating equipment.

3.2.6. Operating Air Compressed System (IA & SA Supplying)

3.2.6.1. IA system delivers clean, dry, and oil-free air to pneumatic control devices, actuators, and instrumentation panels, ensuring reliable operation of critical automation and safety systems.

3.2.6.2. SA system provides compressed air for general utility purposes, including cleaning, cooling, and operating auxiliary equipment or pneumatic tools.

3.2.6.3. Piping routes begin at the air compressor station, which houses compressors, dryers, filters, and storage receivers, and extend through main distribution headers to IA and SA networks.

3.2.6.4. IA piping is routed along pipe racks or in enclosed trenches to critical equipment such as gas turbine skids, control rooms, and pneumatic actuators, while SA piping serves auxiliary areas, workshops, and plant service points. Branch lines for both systems include isolation valves, pressure regulators, and drains to facilitate maintenance and prevent moisture accumulation, while instrumentation such as pressure and flow gauges ensures continuous performance monitoring.

3.2.6.5. Before Power Receiving, IA piping must be completed, pressure-tested, and properly purged to prevent foreign material ingress, ensuring that all pneumatic instruments and control valves operate correctly.

3.2.7. Operating WT/WWT

3.2.7.1. Water Treatment (WT) system delivers clean, treated water, including demineralized or

chemically conditioned water, to boilers, turbines, cooling systems, and auxiliary equipment, ensuring safe and efficient operation. Its piping network transports water from the treatment plant to the necessary equipment while maintaining proper flow, pressure, and water quality, preventing scaling, corrosion, and contamination.

- 3.2.7.2. Wastewater Treatment (WWT) system collects and conveys wastewater from plant operations, including blowdown, condensate return, and process effluents, to treatment units for safe handling or reuse. WWT piping ensures proper drainage, supports hygiene, prevents flooding, and enables water recycling where applicable.
- 3.2.7.3. WT piping begins at treatment units and routes main headers to storage tanks, then distributes through branch lines to HRSG feedwater systems, the deaerator (where provided), cooling towers, and auxiliary equipment. Piping is installed on pipe racks, in trenches, or underground, with proper supports, access points, and insulation for maintenance and operational reliability.
- 3.2.7.4. WWT return lines collect water from equipment, condensers, and process areas, directing it to sumps, manholes, and treatment facilities. Branch lines and valves allow isolation for maintenance, while proper slopes prevent stagnant water and ensure efficient drainage.
- 3.2.7.5. Both WT and WWT piping networks integrate instrumentation, including flow meters, pressure gauges, and level transmitters, to continuously monitor operational parameters.

3.2.8. Demi. Water Production

- 3.2.8.1. Demi. water production is a critical milestone, serving as the primary source for subsequent commissioning activities in the combined cycle power plant.
- 3.2.8.2. Demi. water is essential for hydrostatic testing and chemical cleaning of the HRSG and BOP steam, feedwater, and condensate systems.
- 3.2.8.3. Piping network from the water treatment building through the demi. water tank, pumps, and onward to the HRSG building must be fully installed, supported, and hydro-tested prior to chemical cleaning operations.
- 3.2.8.4. The system comprises raw water intake piping, treatment units such as reverse osmosis (RO) and ion exchange units, storage tanks, pumps, and distribution piping. Demi. water piping transports purified water from the treatment plant to storage tanks and onward to end-use systems under controlled flow and pressure.
- 3.2.8.5. Installation begins at the raw water intake, with main supply piping connecting to the treatment plant, where water is purified through RO and ion exchange units. Treated water is then pumped and filtered into storage tanks, from which main distribution headers deliver demi. water to various plant systems, including boiler feedwater, turbine auxiliaries, and

chemical dosing points. Branch lines extend from the main headers to individual end-use points, following pipe racks, trenches, or underground routes according to plant layout.

3.2.9. CCWP, MCWP, ACWP Load Test / Cooling Tower System Commissioning

- 3.2.9.1. Closed Cooling Water Pump (CCWP), Main Cooling Water Pump (MCWP), and Auxiliary Cooling Water Pump (ACWP) load test are essential for plant operation, and during commissioning, their piping networks undergo flushing, hydrotesting, and functional load testing to ensure integrity and readiness.
- 3.2.9.2. MCW system circulates large volumes of water to the steam turbine condenser, transferring rejected heat to the cooling tower, while the ACW system supplies cooling to auxiliary components such as lube oil coolers, generator air coolers, and heat exchangers.
- 3.2.9.3. CCW system provides a closed-loop network for sensitive equipment, isolating them from open-loop water to minimize fouling and corrosion.
- 3.2.9.4. MCW system piping routes water from the cooling tower basin to pump suction headers, then to condenser water boxes, and back to the cooling tower, while ACW system piping serves auxiliary equipment across the plant, and CCW system piping forms closed loops connected to heat exchangers.
- 3.2.9.5. All cooling water piping is installed on pipe racks, in underground, and pump houses, with supports, guides, and anchors to accommodate hydraulic loads and thermal expansion. Internally, pipes are rubber lined or coated to resist corrosion and scaling, and strainers and filters prevent debris from entering pumps and heat exchangers. Instrumentation, including pressure gauges, flow meters, and temperature sensors, enables real-time performance monitoring, while hydrotesting and chemical cleaning verify integrity and remove construction debris.
- 3.2.9.6. Expansion joints installed after pumps are secured with tie-rods to prevent excessive movement, and large-bore piping, especially GRP, requires careful alignment, elevation checks, and proper backfilling to avoid settlement or offset issues.

3.2.10. HRSG Hydro Test

- 3.2.10.1. Hydro test of the HRSG is a critical verification activity performed to confirm the mechanical integrity and leak-tightness of all pressure parts and associated piping prior to commissioning.
- 3.2.10.2. During the test, steam and water circuits are completely filled with demi. water, pressurized gradually to the specified hydro test pressure, and maintained for the duration required by applicable codes and standards to detect leaks or weaknesses in welds, joints, and pipe materials.
- 3.2.10.3. Temporary manifolder, filling lines, and vent lines are installed, with high-point vents and low-

point drains ensuring complete air removal and controlled draining. The test covers all HP, IP, and LP steam circuits, economizer, evaporator, and superheater sections, with each pressure part fully isolated according to design specifications.

- 3.2.10.4. Special attention is given to cleanliness, support stability, and the installation of temporary blinds, safety valves, and calibrated pressure gauges to prevent over pressurization and ensure accurate monitoring. Steam piping with special materials, such as Alloy P91, follows strict welding procedures and uses transition pieces where necessary to avoid material incompatibility.
- 3.2.10.5. For hydro testing, check valves at stop valve locations are temporarily removed and restored after the test, while blind flanges are installed correctly to withstand test pressures without direct fillet welding.
- 3.2.10.6. Feedwater system piping undergoes hydro testing in coordination with the HRSG steam circuits to allow subsequent chemical cleaning, with boiler feed pump installation completed beforehand.
- 3.2.10.7. Pressure is carefully monitored throughout the test, and all welds, joints, and supports are visually inspected for deformation or leakage.
- 3.2.10.8. Upon successful completion, the system is depressurized in a controlled manner, drained, and dried, followed by cold setting of pipe supports to their final operating positions, confirming readiness for plant startup.

3.2.11. CEP Load Test / CPH Load Test / BFP Load Test

- 3.2.11.1. During the load testing of the Condensate Extraction Pump (CEP), Condensate Pre-Heater (CPH), and Boiler Feed Pump (BFP) systems, all associated piping is installed and commissioned to verify stable and reliable operation of the water-steam cycle.
- 3.2.11.2. CEP piping connects the condenser hot well to the low-pressure feedwater system, with careful alignment to minimize vibration and cavitation, ensuring smooth suction and discharge flow to condensate polishing and heating systems.
- 3.2.11.3. CPH piping routes condensate through the preheater heat exchanger (or the HRSG LP economizer section) to absorb heat from the gas turbine exhaust or a designated heat source, raising condensate temperature, improving cycle efficiency, and preventing thermal shock to downstream equipment. This piping is provided with vents, drains, and bypass lines to facilitate testing, maintenance, and safe operation.
- 3.2.11.4. BFP piping connects the deaerator (or the LP drum where the deaerating function is integral to the HRSG) or feedwater storage tanks to the HRSG feedwater inlet headers, designed for high-pressure delivery with leak-tight joints in compliance with applicable codes. Expansion

joints, control valves, and non-return valves are installed to protect pump operation and prevent backflow.

3.2.11.5. Integration of CEP, CPH, and BFP piping requires precise hydraulic coordination, ensuring stable flow, proper alignment, and reliable pump performance.

3.2.11.6. All welds, joints, and supports are inspected and tested before the load test to prevent leaks or operational failures.

3.2.11.7. Successful completion confirms readiness of the water-steam cycle for subsequent chemical cleaning, commissioning, and operational startup.

3.2.12. HRSG Steam / Feed Water / Condensate Water Chemical Cleaning

3.2.12.1. Chemical cleaning is performed after the CEP, CPH, and BFP load tests and the condensate and feedwater flushing, so that the permanent pumps and lines are proven and gross construction debris has already been removed before chemical circulation begins.

3.2.12.2. Chemical cleaning of the HRSG steam, feedwater, and condensate water systems is a critical pre-commissioning activity aimed at removing construction debris, oils, welding slag, rust, and mill scale from internal surfaces.

3.2.12.3. The process circulates controlled chemical solutions through the HRSG pressure parts and associated piping to ensure all systems are free from contaminants that could cause corrosion, fouling, or equipment damage.

3.2.12.4. By removing foreign materials and passivating metal surfaces, chemical cleaning enhances long-term reliability, prevents early tube failures, and establishes a clean baseline for the water-steam cycle.

3.2.12.5. Temporary piping loops are installed to connect the chemical tanks to the HRSG steam, feedwater, and condensate systems, with vents and drains provided at strategic high and low points for air removal and complete drainage. Isolation blinds are used to protect systems not included in the cleaning loop, and materials compatible with cleaning chemicals, typically stainless steel or lined piping, are applied for corrosion resistance.

3.2.12.6. Proper supports and restraints are installed on temporary lines to prevent vibration or accidental disconnection, and safety devices such as relief valves and spill trays ensure operator and equipment protection.

3.2.12.7. Cleaning procedure begins with hydro-flushing and air removal, followed by circulation of acid or alkaline solutions containing corrosion inhibitors and detergents for a defined period, with continuous monitoring to verify effectiveness.

3.2.12.8. After circulation, spent chemicals are collected and neutralized, and the system is thoroughly rinsed with demineralized water until water chemistry meets target parameters. Finally, the

cleaned piping is dried using compressed air or nitrogen to prevent flash rusting before returning to service.

3.2.12.9. Coordination with chemical cleaning contractors ensures that piping installation and hydro testing for HRSG and fuel system lines are completed in advance to allow timely cleaning and compliance with GT supplier requirements.

3.2.12.10. Chemical cleaning of HRSG internal piping is typically performed by specialized subcontractors. Hydrotest planning must take into account the mobilization schedule of the chemical cleaning team and the connection plans of the HRSG piping to ensure an uninterrupted cleaning and testing sequence.

3.2.12.11. Acid-based chemical cleaning is generally applied to steam, feedwater, and fuel supply systems, while alkali-based cleaning is used for condensate water systems. For steam piping, chemical cleaning is not always mandatory, but it is applied to reduce the duration of steam blowing and to secure sufficient commissioning time.

3.2.13. Fuel Supply Systems (Fuel Oil or Fuel Gas)

3.2.13.1. Fuel supply system is responsible for transporting fuel from the external supply source to the gas turbine fuel skids and associated equipment. Depending on the design of the plant, the system may handle natural gas, fuel oil, or dual-fuel operation.

3.2.13.2. Piping system ensures that the gas turbine receives fuel at the required pressure, temperature, and flow rate for stable and efficient combustion.

3.2.13.3. The primary function of the fuel supply piping is to guarantee a safe, continuous, and reliable flow of fuel to the gas turbines. In the case of natural gas, the piping system delivers the gas from the external supplier or pipeline network to the gas receiving station, filtration units, and fuel conditioning skid before entering the turbine combustors.

3.2.13.4. For fuel oil operation, the system transfers oil from the storage tanks through treatment facilities and forwarding pumps to the turbine fuel skids. The system also incorporates pressure regulating devices, filtration units, and safety interlocks to maintain fuel quality and operational safety.

3.2.13.5. The installation route of the fuel supply piping begins at the plant boundary, where the external fuel connection point or supply station is located.

3.2.13.6. For liquid fuel, pipelines are routed from the main fuel oil storage tanks to the fuel forwarding pumps, through oil treatment shelters, and finally to the turbine fuel skids.

3.2.13.7. For natural gas, underground or aboveground pipelines run from the gas receiving station to the gas conditioning skid, typically installed near the turbine building.

3.2.13.8. Within the plant, the fuel supply piping is generally installed on pipe racks and routed along

designated utility corridors to minimize congestion and ensure accessibility for inspection and maintenance. Specialized connections are also provided for emergency shutdown valves and fire safety systems along the fuel piping route.

3.2.13.9. Fuel supply system generally uses carbon steel pipes to handle the specific characteristics of natural gas or fuel oil. Stainless steel materials are used in areas where corrosion resistance is critical.

3.2.13.10. The system includes high-pressure fuel gas pipelines, low-pressure distribution lines, fuel oil transfer lines, forwarding lines, and return lines for circulation. Valves, filters, pressure regulators, and safety relief devices are installed at key points along the piping to maintain operational stability and protect the turbines.

3.2.13.11. Additionally, instrumentation piping for pressure, flow, and temperature monitoring is integrated into the fuel system to ensure precise control during turbine operation.

3.2.13.12. This system is essential for plant startup. A pig cleaning or acid chemical cleaning procedure is mandatory to clean the pipe internals after hydro testing. To avoid delays, prioritize the installation of piping and supports simultaneously, and schedule hydro tests as early as possible.

3.2.13.13. After cleaning, the pipes should be air-blown and then nitrogen-purged to prevent internal corrosion before fuel is introduced.

3.2.13.14. Fuel gas or fuel oil is admitted to the GT only after hydro testing, cleaning, nitrogen purging, and leak tightness verification of the entire fuel supply line are complete, which is a prerequisite for GT initial firing.

3.2.14. GT System Functional Test / GT Lube Oil Flushing

3.2.14.1. During the GT system functional test and lube oil flushing stages, all associated piping is installed and commissioned to ensure operational reliability of the turbine and auxiliary systems.

3.2.14.2. Piping networks include fuel gas and fuel oil supply lines, compressor inlet air, exhaust ducts, and auxiliary connections such as sealing, cooling, and starting air lines. All piping is installed to precise tolerances with appropriate supports, expansion joints, and insulation to accommodate thermal expansion, vibration, and high-temperature conditions.

3.2.14.3. GT lube oil system piping connects reservoirs, pumps, coolers, filters, and bearings, with careful alignment of bends, slopes, and drains to prevent stagnation or air entrapment. Welded and flanged joints are inspected and pressure-tested to ensure leak-tightness before flushing.

3.2.14.4. During lube oil flushing, high-velocity circulation removes construction debris, rust, and contaminants, supported by temporary lines, bypasses, return headers, filters, and strainers to

maintain turbulent flow.

3.2.14.5. Cooling water, instrument air, and drain lines are installed in parallel to support stable operation and effective lubrication. Fuel system piping is chemically cleaned and preserved with nitrogen, with filters verified at the GT inlet to prevent damage and minimize vendor liability risks.

3.2.14.6. CCW and IA lines to the generator, purge air cooler, lube oil cooler, and fuel gas heat exchanger must be completed prior to turbine startup. All piping, supports, and pressure tests for CCW and IA systems are finalized before sequential GT operation, with additional isolation valves installed on branch lines if schedule constraints prevent full completion.

3.2.15. GT Initial Firing / GT Synchronization

3.2.15.1. GT Initial Firing requires all fuel and auxiliary piping systems to be fully installed and verified for integrity to support safe turbine ignition.

3.2.15.2. Fuel gas supply line must be complete from the main receiving station through conditioning skids, including filters, heaters, and pressure controls, to the GT fuel gas manifold. Where liquid fuel capability exists, fuel oil piping from storage tanks, treatment systems, and pumps to the GT nozzles must also be commissioned. Instrument air and nitrogen piping are essential for valve actuation, purging, and safety interlocks, while ignition gas and starting air lines ensure turbine acceleration to firing speed.

3.2.15.3. Lube oil and associated cooling water circuits must be operational to protect bearings, and seal air and anti-icing lines must be ready for reliable operation under varying conditions.

3.2.15.4. Exhaust piping, including expansion joints, silencers, and stacks, must be properly aligned, insulated, and supported to safely discharge high-temperature combustion gases.

3.2.15.5. During GT synchronization, generator and turbine cooling water networks must be in service to support stable operation.

3.2.15.6. Instrument air lines are critical for maintaining proper control, while drain, vent, and firefighting water systems ensure safe and protected startup conditions.

3.2.15.7. All piping systems must undergo hydro-testing, pressure testing, flushing, and commissioning prior to initial firing. Full readiness of fuel, lube oil, cooling, air, exhaust, and auxiliary lines ensures a smooth transition from initial firing to grid connection and stable commercial operation.

3.2.16. Steam Blowing

3.2.16.1. Steam Blowing is a critical commissioning activity conducted to remove construction debris, welding slag, rust, oil, and other foreign particles from the steam piping prior to turbine admission.

- 3.2.16.2. High-pressure steam piping from the HRSG to temporary blow-out connections must be fully installed, hydro-tested, and routed to safe discharge points with proper strength, expansion allowances, and silencers.
- 3.2.16.3. Permanent steam piping, including main and reheated lines, must have all welds inspected, stress-relieved if required, and supported with hangers and expansion joints to maintain alignment during thermal cycling.
- 3.2.16.4. Steam bypass systems divert steam safely to the condenser or atmosphere and must be tested for tightness and operability.
- 3.2.16.5. Auxiliary piping, such as demineralized water supply, condensate return, and drain and vent lines, is required to generate steam, return condensate, and remove air from the system. Instrument air piping must be completed to operate blow-out, bypass, and safety valves, while nitrogen and chemical dosing piping ensure proper system conditioning.
- 3.2.16.6. Temporary silencers and exhaust stacks must be securely installed to handle vibration, thrust, and high-pressure steam discharge during the process.
- 3.2.16.7. All piping systems must be fully installed, pressure-tested, and commissioned prior to steam blowing to prevent turbine damage and ensure operational safety.
- 3.2.16.8. Installation of temporary piping and supports requires careful planning, often taking several months, and all temporary lines must be removed before admitting live steam.
- 3.2.16.9. Proper coordination of main, bypass, auxiliary, and temporary systems ensures that steam blowing is executed efficiently, safely, and without impact on downstream equipment.
- 3.2.16.10. Steam blowing can only be performed after hydro testing and insulation of steam and bypass steam piping is completed. Therefore, hydro testing of all steam and bypass piping must be completed at least three months prior to the scheduled steam blowing period.

3.2.17. ST Lube Oil Flushing

- 3.2.17.1. ST Lube Oil Flushing is a critical pre-commissioning activity aimed at removing construction debris, welding residues, and contaminants from the lubrication system to protect turbine bearings and rotors.
- 3.2.17.2. Lube oil piping system must be fully installed, connecting the reservoir to main, journal, and thrust bearings, as well as auxiliary systems such as coolers, filters, and pumps. Piping routing is designed to maintain proper flow and slope to prevent air pockets, oil stagnation, or uneven distribution during flushing. All welded joints, flanges, and fittings are inspected and hydrotested to ensure leak-tight integrity.
- 3.2.17.3. Cooling water piping for the lube oil coolers must be operational to maintain oil temperature within design limits. Temporary flushing lines and bypass piping are installed to allow high-

velocity circulation, while strategically placed drains and vents remove trapped air and old oil.

3.2.17.4. Filters and strainers are integrated to capture debris during circulation. Instrumentation and control piping, including pressure, flow, and temperature sensors, must be completed to monitor the flushing process in real time.

3.2.17.5. Auxiliary lines, such as nitrogen or air for pressurization, and safety devices like relief and shut-off valves, are installed to ensure proper operation and protection of pumps and bearings.

3.2.17.6. After successful flushing, the piping system is drained and prepared for normal turbine lubrication during startup and operation.

3.2.18. Steam Turbine Steam Admission

3.2.18.1. ST Steam Admission is a critical commissioning activity designed to verify turbine operability, system integrity, and the effectiveness of chemical conditioning systems.

3.2.18.2. Main steam piping from the HRSG outlet headers to the turbine inlet, including HP, IP, and LP lines, must be fully installed, aligned, and hydro-tested. Piping supports, anchors, and expansion joints are provided to accommodate thermal expansion, vibration, and pressure surges during steam introduction.

3.2.18.3. Bypass steam lines are installed to safely divert steam to the condenser or blow-off system, with valves, silencers, and temporary stacks managing pressure and temperature. Drain and vent piping at strategic high and low points remove condensate, air, and debris, ensuring uniform heating and preventing water hammer or thermal shock.

3.2.18.4. Sampling and chemical dosing lines are connected upstream and downstream of the turbine to monitor steam and condensate chemistry and inject inhibitors, oxygen scavengers, or corrosion control agents. Isolation valves, back-pressure regulators, and drains are integrated into dosing and sampling lines for safe and controlled operation.

3.2.18.5. Auxiliary piping, including seal steam, lube oil cooling, and instrument air lines, is installed to maintain turbine shaft sealing, bearing protection, and real-time monitoring.

3.2.18.6. During commissioning, operators gradually open control and stop valves, continuously observing pressure, temperature, and chemical parameters while ensuring proper venting, drainage, and chemical dosing.

3.2.18.7. Once steam is admitted, STG building becomes a hot commissioning zone, with remaining piping work suspended and technical advisors monitoring spring supports and snubbers for correct operation.

3.2.18.8. During the Steam Admission (Hot function) period, the support supplier dispatches a Technical Advisor (TA) to confirm that piping movement aligns with the designed thermal expansion directions. If discrepancies are identified, design improvements or corrective adjustments are

implemented to ensure proper thermal expansion accommodation and safe operation of the steam system.

3.2.19. Performance Test

- 3.2.19.1. Performance Test stage in a combined cycle power plant is conducted to verify that all plant systems, including the steam, gas turbine, and auxiliary piping networks, operate in accordance with design specifications under full load conditions.
- 3.2.19.2. Performance testing is carried out in accordance with the contract and the applicable test code (e.g., ASME PTC 46 for overall plant performance) after the plant has reached stable full-load operation.
- 3.2.19.3. All permanent piping, insulation, and supports must be complete, and temporary strainers, blinds, and test spools must be removed and the systems restored to their permanent configuration.
- 3.2.19.4. Test boundaries require isolation of extraneous flows. Isolation valves, drains, vents, bypass, and blowdown valves within the test boundary must be verified as tight shut-off, since leakage past valves affects the heat balance and the measured output and efficiency.
- 3.2.19.5. Measurement points and test connections, such as the fuel gas flow meter, feedwater flow element, and steam and condensate temperature and pressure taps, must be installed and calibrated before the test.
- 3.2.19.6. Fuel, demineralized water, cooling water, and instrument air supply lines must be capable of continuous full-load operation throughout the test.

3.2.20. Reliability Test

- 3.2.20.1. Reliability Test stage in a combined cycle power plant is conducted to verify that all installed piping systems and connected equipment can operate continuously and safely under sustained plant operating conditions.
- 3.2.20.2. During the reliability run, the plant operates continuously for the duration specified in the contract without unplanned outages.
- 3.2.20.3. Piping is monitored for leakage at flanges, valves, and expansion joints, abnormal vibration or water hammer in steam, condensate, and drain lines, and abnormal movement of pipe supports.
- 3.2.20.4. After the first hot operation, a hot walk-down verifies that spring hangers are within their hot load range, guides and sliding supports move freely, and no interference or rubbing occurs. Deficiencies are corrected under the guidance of the supplier TA before or during the reliability run.

3.2.20.5. Remaining punch items that affect continuous operation, such as missing insulation, drain and vent modifications, and support corrections, must be closed out without interrupting the run, using planned outage windows if required.

3.2.20.6. Piping-related trips or forced outages during the run must be analyzed and corrected, and the run restarted in accordance with the contract.

3.2.21. COD (Commercial Operation Date)

3.2.21.1. Commercial Operation Date (COD) marks the transition of the combined cycle power plant from commissioning to full commercial operation. During this stage, all piping systems must be fully installed, tested, and ready for continuous operation under rated plant conditions.

3.2.21.2. All remaining punch list items, NCRs (Non-Conformance Reports), and SORs must be resolved before COD. This includes essential final tasks such as touch-up painting, insulation installation, labeling, and the demolition of scaffolding. It also includes correcting any errors in support installation (e.g., missing guides or incomplete welds).

4. Piping Construction Management

4.1. Piping construction in combined cycle power plants is a critical activity that impacts project schedule, quality, safety, and reliable plant operation. Effective management requires a structured approach integrating contract requirements, technical specifications, material control, manpower, and commissioning priorities.

4.2. General Management

4.2.1. Prior to commencing piping works, review contract documents and technical specifications, particularly ASME B31.1 Power Piping Code, to ensure compliance in design, material selection, welding, NDE, PWHT, and testing. Review the overall project schedule to align piping construction with key commissioning milestones such as power receiving, initial GT operation, and steam blowing.

4.2.2. All stages of piping work—including design verification, material selection, fabrication, welding, installation, and testing—must comply fully with code requirements. The design stage should confirm that piping stress analysis, allowable loads, thermal expansion, and vibration limits are within acceptable ranges. Material selection must consider pressure rating, temperature, corrosion resistance, and compatibility with fluids, as defined in the project specifications. Fabrication and installation must follow approved procedures to prevent defects, misalignment, or mechanical interference.

4.2.3. All welding activities must adhere strictly to approved Welding Procedure Specifications (WPS), with supporting Procedure Qualification Records (PQR) demonstrating that the procedure can produce acceptable welds.

- 4.2.4. Welders performing work must be qualified through Welder Qualification Tests (WQT) for the specific materials, thicknesses, and welding processes used.
- 4.2.5. For high-alloy or high-temperature materials such as P91, P22, or P11, WPS/PQR/WQT compliance is critical to prevent premature failure of pressure parts during operation.
- 4.2.6. Design parameters must be defined according to the material class, including allowable stress, design pressure, design temperature, and corrosion allowance.
- 4.2.7. Different material classes, such as carbon steel, low-alloy steel, stainless steel, and high-temperature alloys, have specific requirements for wall thickness, joint design, and support spacing. Ensuring correct parameters prevents overstressing, deformation, and potential failure of the piping system during plant operation.
- 4.2.8. All critical welds and components must undergo the required NDE methods, such as radiography, ultrasonic testing, magnetic particle, or dye penetrant inspection, to verify weld integrity and detect defects.
- 4.2.9. Post-Weld Heat Treatment (PWHT) must be applied as specified for material classes susceptible to residual stresses or hardness issues, including alloy steels.
- 4.2.10. Adherence to NDE and PWHT requirements ensures mechanical integrity, reduces the risk of cracking or failure, and complies with ASME B31.1 standards for high-pressure and high-temperature piping systems.

4.3. Drawing Control

- 4.3.1. Maintain continuous coordination with the engineering and design teams to incorporate the latest drawing revisions. Ensure that field installation follows the approved drawings, reflecting any design changes or updates.

4.4. Material Control

- 4.4.1. Establish a comprehensive material management system based on the Bill of Materials from the Design Team. Secure dedicated storage, containers, and laydown areas, with special attention to critical tagged items such as spring supports and snubbers. Track material issuance and inventory in real time. Early procurement of long-lead items, including alloy piping (e.g., P91, P22), welding electrodes, and WQT test coupons, is essential to prevent delays.
- 4.4.2. Support materials are delivered as a mix of bulk and tagged items. After material receiving inspection, segregate and manage tagged items separately, with particular attention to critical items such as spring supports and snubbers.

4.5. Welding Test Coupon and Welding Rod

- 4.5.1. Confirm the availability of Test Coupon pipes for Welder Qualification. For alloy systems such

as P91, P22, and P11, which are considered long-term delivery items, coordinate with the Procurement Team in advance to ensure timely supply.

- 4.5.2. For welding electrodes, check the contractual requirements for the supplier, estimate the expected quantities, and coordinate with the Procurement Team to ensure timely delivery.

4.6. Manpower Mobilization

- 4.6.1. When reviewing the manpower mobilization plan, secure resources for equipment, scaffolding, hydro test teams, and common tools. Scaffolding support is essential for smooth piping construction, so a dedicated scaffolding team must be secured in advance.
- 4.6.2. Assign a support team from the piping installation stage to allow simultaneous execution of piping and support installation. Manage delivery schedules of support materials accordingly.
- 4.6.3. The most challenging installation issue in power plant piping works is the installation of steam system supports. Secure experienced foremen (supervisors) in advance.

4.7. System-Based Construction Priority and Control

- 4.7.1. Prioritize piping installation for systems critical to early commissioning milestones, such as power receiving, demi. water production, and initial GT operation.
- 4.7.2. BOP piping installation should begin in bulk for all available areas as soon as materials and drawings are ready, then finalized sequentially by system according to commissioning schedules.
- 4.7.3. Start with utility systems such as firefighting, potable water, instrument air, and WT/WWT system, followed by installation of various system headers on pipe racks. Once buildings and equipment are installed, proceed with branch piping installation.
- 4.7.4. Large-bore piping, pipe rack headers, and systems with high material volumes such as steam, feedwater, condensate, and circulation water should be installed first in bulk, then completed sequentially by system according to commissioning schedules.
- 4.7.5. Carefully review the overall project schedule, as hydro tests become extremely difficult after commissioning begins.

4.8. Hydro Testing and Isolation

- 4.8.1. Plan hydro tests before commissioning wherever possible, as testing after commissioning is challenging. When post-commissioning hydro tests are unavoidable, install isolation valves at the upstream side of each building for operational flexibility. Ensure accessibility for valves and proper routing of vent and drain lines, considering test package requirements and maintenance access.

4.9. Flexible Execution

- 4.9.1. Adjust installation sequencing based on site conditions, such as building and equipment

progress, pipe rack availability, material delivery, and manpower constraints. Maintain flexibility to proceed with bulk piping installation in available areas while coordinating with other trades and commissioning schedules.

4.10. U/G (Underground) Piping Works

- 4.10.1. Underground piping systems in a combined cycle power plant are critical for utilities such as fire water, cooling water, and process water and fuel oil or gas supplying. Proper installation ensures system reliability, prevents leakage, and facilitates maintenance while complying with applicable codes such as ASME B31.1 Power Piping Code, AWWA standards for water systems, and material-specific installation practices.
- 4.10.2. Excavation, bedding, and backfilling must be executed according to project specifications, with adequate compaction to prevent settlement.
- 4.10.3. Piping alignment, grade, and slope must be verified to ensure proper flow, drainage, and hydrostatic testing.
- 4.10.4. All buried piping should include provisions for thermal expansion, access for inspection, and corrosion protection as applicable.

4.10.5. HDPE Piping

- 4.10.5.1. HDPE piping is widely used for fire water, potable water, and service water systems. Joints are typically fused using butt fusion or electrofusion techniques, ensuring homogenous, leak-proof connections.
- 4.10.5.2. The pipe must be laid on well-compacted granular bedding and restrained using anchor blocks at bends, tees, and valves. Backfill should be free of large rocks or debris, with proper compaction to avoid pipe deformation. HDPE piping requires traceability of fusion parameters, including temperature, pressure, and fusion time, and post-installation hydrostatic testing to verify integrity.
- 4.10.5.3. The installation of HDPE underground piping begins with excavation of the trench to the specified depth and width, ensuring stable sidewalls and a firm foundation. A bedding layer of well-graded granular material is placed and compacted to provide uniform support and prevent pipe deformation.
- 4.10.5.4. HDPE pipes are then lowered into the trench using appropriate lifting equipment to avoid damage, maintaining alignment and grade as per design drawings. Jointing is carried out by butt fusion or electrofusion, where pipe ends are prepared, heated, and fused under controlled temperature and pressure to achieve a homogeneous, leak-free joint.
- 4.10.5.5. Each fusion joint is visually inspected and logged to ensure compliance with quality requirements.

- 4.10.5.6. Once jointing is complete, the pipeline is subjected to a hydrostatic pressure test to confirm its integrity before backfilling. Initial backfill is performed with fine granular soil around and above the pipe crown to protect against point loading.
- 4.10.5.7. Successive backfilling is done in layers with mechanical compaction to achieve the required density and prevent settlement. Anchor blocks are constructed at bends, tees, and valve chambers to resist thrust forces during operation.
- 4.10.5.8. Finally, as-built surveys are performed, and installation records including fusion logs, test results, and material certificates are submitted for project documentation and turnover.
- 4.10.5.9. HDPE piping for fire-fighting systems must comply with NFPA and AWWA standards. If AWWA standards are not applied, FM Approved or UL Listed materials are required. Fusion welding machines for butt and socket fusion must be available for each pipe size. Potable water piping also uses HDPE material.

4.10.6. GRP Piping

- 4.10.6.1. GRP piping is used for circulating water and chemical-resistant applications. GRP pipes should be installed on smooth, uniform bedding to prevent point loads and potential cracking. Jointing is performed using flanged couplings, adhesive bonding, or mechanical joint systems as per manufacturer recommendations.
- 4.10.6.2. GRP piping should include thrust blocks, and corrosion-resistant anchors at changes in direction. Pressure testing must follow project-specific hydrotest procedures, taking care to maintain uniform support and prevent vibration.
- 4.10.6.3. The installation of GRP underground piping begins with excavation of the trench to the specified dimensions, ensuring a stable foundation free from sharp objects or debris that may damage the pipe surface. A bedding layer of compacted sand or fine granular material is prepared to provide uniform support and distribute the load evenly.
- 4.10.6.4. GRP pipes are then carefully lowered into the trench using slings and lifting equipment, avoiding impact or concentrated stresses on the pipe body. Pipe joints are typically assembled using adhesive-bonded sleeve couplings or mechanical joints, which must be cleaned, aligned, and cured under controlled conditions to achieve leak-tight performance.
- 4.10.6.5. Once jointing is complete, alignment and grade are verified to ensure proper flow characteristics and prevent localized stresses. Hydrostatic testing is performed to confirm the structural and leak integrity of the system before backfilling.
- 4.10.6.6. Initial backfill is applied with selected granular material up to the pipe crown, compacted uniformly around the sides to provide lateral support. Subsequent backfill layers are placed

and compacted in stages to restore the trench to its original grade while protecting the pipe from external loading.

4.10.6.7. Thrust blocks or restrained joints are installed at changes in direction and major fittings to resist hydraulic forces during operation. Finally, an as-built survey is completed, and all test reports and quality records are compiled for system turnover.

4.10.6.8. During backfilling, pipe sagging may occur. Pipes should be secured using chain blocks or similar tools before covering. Alignment deviations between U/G and A/G piping are common; therefore, coordinate elevations and offsets with design drawings before connection.

4.10.7. CPVC Piping

4.10.7.1. CPVC piping is typically applied for chemical service water underground systems. CPVC joints are primarily solvent-welded and must be installed at ambient temperatures to ensure chemical compatibility and leak-free connections.

4.10.7.2. Pipes must be laid on sand or fine granular bedding to prevent point loading and thermal stress cracking. Underground routing should avoid sharp bends, and expansion loops or offsets should be incorporated to accommodate thermal expansion. Hydro testing is mandatory after installation to validate system integrity before backfilling.

4.10.7.3. The construction of CPVC underground piping begins with trench excavation according to design depth and width, ensuring a uniform grade to avoid stress points. A bedding layer of fine sand or granular material is compacted to provide continuous pipe support and prevent point loading. Pipes are lowered into the trench with care to avoid scratches or impact damage that could compromise chemical resistance.

4.10.7.4. Joints are typically solvent-cemented, requiring surface preparation, proper application of primer and cement, and adequate curing time before pressurization. Alignment and slope are checked to maintain hydraulic efficiency and prevent air pockets or low points.

4.10.7.5. Once jointing is completed, hydrostatic testing is performed to verify the leak-tightness of the piping system. Initial backfill with soft, non-cohesive material is placed and compacted carefully up to the pipe crown to stabilize the pipeline.

4.10.7.6. Further layers of backfill are added and compacted in stages to protect the pipe from settlement and external loads. Thrust restraint, usually by mechanical anchoring or encasement, is provided at directional changes and fittings. Finally, system integrity records and as-built drawings are reviewed and submitted for commissioning readiness.

4.10.8. PE Coated Steel Piping

4.10.8.1. PE-coated piping is commonly used for high-pressure water, fuel, or process lines requiring

corrosion protection. The steel pipe is jointed using flanges or welded connections, followed by the application of field-applied polyethylene coating or heat-shrink sleeves at welded areas.

- 4.10.8.2. Cathodic protection may be installed for additional corrosion prevention. Bedding and backfilling must be performed carefully to avoid damaging the coating. Anchoring and thrust blocks are required at tees, bends, and valves to resist hydraulic thrust. Hydrostatic testing of steel piping is conducted before backfilling, and all coating integrity must be verified post-test.
- 4.10.8.3. The installation of PE-coated steel underground piping begins with excavation of the trench to the required depth, considering both pipe diameter and protection cover requirements.
- 4.10.8.4. The trench bottom is prepared with a bedding layer of sand or fine gravel to provide uniform support and avoid coating damage. Pipes are carefully lowered using slings or padded lifting devices to prevent scratching or peeling of the external polyethylene coating.
- 4.10.8.5. Welded joints are commonly applied, and after welding, field joint coating is applied using heat-shrink sleeves or liquid epoxy to restore corrosion protection. NDE is conducted on welds to verify joint integrity.
- 4.10.8.6. Once joint coating is completed, holiday detection tests are performed to confirm coating continuity and prevent corrosion risk.
- 4.10.8.7. Hydro testing is carried out to ensure pressure integrity before burial. Initial backfilling with fine granular material is compacted around the pipe to protect the coating and provide side support. Successive backfill layers are placed and compacted until the trench is restored to grade, with caution to avoid rocks or hard material contacting the coating. Finally, cathodic protection systems are installed and commissioned to provide long-term corrosion resistance of the buried steel pipeline.
- 4.10.8.8. Compressed air distribution piping (if required) is usually stainless steel with PE coating. Weld quality and joint integrity are critical due to cyclic pressures. Coating inspection and holiday testing must be performed before backfilling.
- 4.10.8.9. Drainage points (Air trap) must be provided at low elevations to prevent condensate accumulation. During backfilling, coating damage must be avoided, and cathodic protection considered depending on soil resistivity.
- 4.10.9. When underground piping connects to aboveground systems without isolation valves, pressure tests for A/G piping must include the U/G sections. Therefore, isolation valve installation must always be confirmed in advance.
- 4.10.10. U/G piping installed early is at risk of damage during subsequent construction. To mitigate this, pipelines should be protected with sand bedding and covered with protective boards. This

requirement must be reflected in the civil design.

4.10.11. Most utility underground piping connects to storage tanks. The tank installation schedule must be confirmed, and piping works planned to be completed before or immediately after tank erection.

4.11. A/G (Above-Ground) Piping Works

4.11.1. The construction of aboveground piping systems in combined cycle power plants requires strict adherence to applicable standards such as ASME B31.1 Power Piping Code and project-specific specifications. Prior to installation, all piping materials, weld fittings, flanges, and supports must be verified against approved drawings and material certificates. Piping routes are laid out in accordance with isometric drawings and general arrangement plans, ensuring proper alignment, accessibility, and clearance from adjacent structures and equipment.

4.11.2. The installation of above ground piping begins with the completion of structural steel and pipe rack erection to provide stable pipe support. Major headers are installed first, followed by branch lines connected to equipment nozzles. Pipe supports and anchors are then positioned to ensure proper alignment and thermal expansion allowance.

4.11.3. Pipe spools are fabricated in the workshop or field following approved Welding Procedure Specifications (WPS) and are subjected to dimensional checks before erection. During lifting and placement, suitable rigging methods are used to prevent deformation or coating damage, particularly for large-diameter and alloy steel pipes. Alignment at joints is checked with precision tools to ensure compliance with tolerances, and tack welding is performed before final welding. All welding operations are carried out by qualified welders and subsequently inspected by NDE to confirm quality and code compliance.

4.11.4. Pipe supports, hangers, guides, and anchors are installed progressively as piping is erected, following the support drawings and manufacturer's load data. Spring hangers and snubbers are adjusted to maintain proper thermal movement and to absorb vibration during operation. Where thermal expansion is significant, expansion joints or loops are installed in accordance with the stress analysis report. After erection, piping systems are cleaned, flushed, or blown out to remove construction debris and contaminants.

4.11.5. Hydrostatic or pneumatic pressure testing is performed in designated test packages, ensuring that the piping system meets code requirements for strength and leak tightness. Upon successful testing, insulation and painting works are carried out as per specification to provide thermal protection and corrosion resistance. Finally, as-built documentation is prepared, and the piping system is handed over for commissioning and operational readiness.

4.11.6. Alloy Steel Piping (P91, P22, P11)

- 4.13.1.1. Alloy steel piping installation begins with pre-inspection of materials to confirm correct heat numbers and mechanical properties. Spools are handled with caution to prevent damage at bevel ends.
- 4.13.1.2. Before welding, preheating is applied, and temperature is monitored continuously during the welding process. Welding is carried out strictly in accordance with approved WPS, with inter pass temperatures controlled.
- 4.13.1.3. After PWHT is conducted on designated joints, using calibrated heating equipment. NDE methods such as radiography or ultrasonic testing are applied to critical welds to confirm integrity.
- 4.13.1.4. During erection, expansion loops, guides, and anchors are installed to accommodate thermal movement. Stress analysis recommendations are followed closely to avoid overstress on high-temperature lines.
- 4.13.1.5. Once the system is erected, hydrostatic testing is performed under controlled conditions. The system is then heated up and cooled down at controlled rates during commissioning to limit thermal stress before operation.
- 4.11.7. Carbon Steel Piping
 - 4.11.7.1. Carbon steel piping installation begins with the inspection of fabricated spools to ensure dimensional accuracy and weld quality. The spools are then transported to the erection area using lifting equipment with soft slings to prevent surface damage.
 - 4.11.7.2. Before erection, supports such as guides, shoes, and anchors are installed in line with design drawings. Spools are aligned and tack-welded in position, followed by full welding performed by qualified welders.
 - 4.11.7.3. Welded joints are visually inspected, and NDE is carried out on selected joints per code requirements. As erection progresses, supports are adjusted to maintain proper slope and alignment.
 - 4.11.7.4. Once the system is complete, hydrostatic testing is performed to verify leak-tightness and strength.
 - 4.11.7.5. After testing, the piping is dried and surface-treated with primer and paint to protect against corrosion. Final alignment and punch list clearance are completed before system turnover.
- 4.11.8. Stainless Steel Piping
 - 4.11.8.1. Stainless steel piping installation starts with proper handling of spools using non-ferrous lifting gear to avoid contamination.
 - 4.11.8.2. Pipes are stored on wooden sleepers or lined racks to prevent contact with carbon steel.

4.11.8.3. During erection, supports are installed with protective liners to prevent galvanic corrosion. Spools are carefully aligned and welded using low-heat input techniques to minimize distortion.

4.11.8.4. After welding, joints undergo dye penetrant testing and dimensional checks. Pickling and passivation are carried out on weld seams and heat-affected zones to restore corrosion resistance.

4.11.8.5. Once installed, the piping is pressure-tested with chloride-free water to prevent pitting. Supports are adjusted to avoid overstress on fittings and flanges.

4.11.8.6. After successful testing, the system is cleaned, dried, and covered until commissioning.

4.11.8.7. Final documentation is prepared to certify compliance with project and code requirements.

4.11.9. Galvanized / Coated Steel Piping

4.11.9.1. Installation of galvanized or coated piping begins with inspection of the coating for scratches, peeling, or defects.

4.11.9.2. Pipes are handled carefully using padded slings to avoid damaging the protective surface. Supports are installed with rubber or plastic liners to protect the coating during operation. Spools are aligned and joined either by flange or threaded connections, with welding minimized to avoid burning the protective layer.

4.11.9.3. Where field welds are necessary, the area is ground, welded, and recoated immediately with approved paint or zinc spray.

4.11.9.4. Threaded joints are sealed with appropriate compounds to ensure leak tightness. After erection, the system undergoes pressure testing to verify integrity.

4.11.9.5. Any coating damages identified during installation or testing are repaired and inspected. The piping is then cleaned, dried, and preserved until turnover.

4.11.9.6. Final inspection ensures compliance with corrosion protection requirements before system handover.

4.12. Steam Piping

4.12.1. Steam piping installation in a combined cycle power plant requires strict adherence to ASME B31.1 Power Piping Code and project specifications, as these systems operate under high temperature and pressure. The process begins with a detailed review of isometric drawings, stress analysis reports, and material traceability records to ensure all components are correct in material grade, thickness, and rating.

4.12.2. Spools and fittings are transported and stored with protective measures to prevent damage to bevel ends and identification markings. Before erection, supports such as guides, anchors,

and spring hangers are installed in accordance with the stress analysis to properly accommodate thermal expansion and prevent overstress.

- 4.12.3. During alignment, piping is set to exact dimensions and slopes, and tolerances at turbine or HRSG connections are verified using precision tools.
- 4.12.4. Welding is performed only by qualified welders in accordance with approved WPS, with preheating, inter pass temperature control, and post-weld heat treatment (PWHT) applied as required for alloy steels such as P91, P22, and P11.
- 4.12.5. Each weld undergoes rigorous NDE, including radiographic or ultrasonic testing, with acceptance criteria per ASME standards.
- 4.12.6. After welding, dimensional re-checks are performed to confirm that expansion loops, offsets, and hanger locations are in compliance with stress design.
- 4.12.7. Hydrostatic testing of steam piping is executed before insulation and with calibrated instruments, ensuring proper pressurization, venting, and drainage to avoid water hammer or pressure surges.
- 4.12.8. Once testing is complete, the system is dried thoroughly to prevent corrosion and internal oxidation. Insulation and cladding are then installed with appropriate sealing to minimize heat loss and protect personnel.
- 4.12.9. Critical Inspection and Verification Requirements
 - 4.12.9.1. Material Verification – Confirm all pipes, fittings, and valves match the required grade (e.g., P91, P22), wall thickness, and pressure class; verify heat numbers and MTRs.
 - 4.12.9.2. Dimensional and Alignment Checks – Ensure correct slope, spacing, and alignment, particularly at turbine, HRSG, and valve connections, to prevent stress concentration.
 - 4.12.9.3. Welding Quality Control – Verify WPS/PQR/WQT compliance, preheat and inter pass temperature monitoring, electrode/buffer rod control, and proper PWHT application.
 - 4.12.9.4. NDE Compliance – Perform RT, UT, MT, or PT on welds per code requirements; document acceptance results before proceeding to the next stage.
 - 4.12.9.5. Support and Hanger Inspection – Confirm installation of spring hangers, guides, anchors, and snubbers per stress analysis; check cold and hot settings for accuracy.
 - 4.12.9.6. Expansion and Flexibility Verification – Ensure expansion loops and offsets are installed as designed to absorb thermal growth without overstressing the system.
 - 4.12.9.7. Hydrostatic Test Preparation – Check that temporary vents, drains, and isolation valves are installed; confirm calibrated gauges and maintain test pressure per code.
 - 4.12.9.8. Post-Test Preservation – Ensure complete draining, drying, and temporary preservation (e.g.,

nitrogen purging) to prevent corrosion prior to steam blowing or commissioning.

4.12.9.9. Final Documentation – Verify inspection records, NDE reports, PWHT charts, and material traceability are complete for turnover.

4.13. Support Installation

- 4.13.1. Properly designed and installed pipe supports are essential to ensure that aboveground piping systems operate safely under thermal expansion, vibration, and sustained operational loads. According to ASME B31.1 Power Piping Code and recognized industry practices, the support system must restrain or guide piping where required, while also permitting controlled movement. Inadequate support installation may lead to overstress, misalignment, or even damage to critical equipment such as turbines, HRSGs, and heat exchangers.
- 4.13.2. During installation, cold pull should be strictly avoided, particularly in the vicinity of turbine connections. Work must commence at the turbine nozzle and extend outward, thereby minimizing stress transfer to sensitive equipment.
- 4.13.3. For high-temperature and high-pressure steam piping, special care must be given to the design and placement of supports. These supports must be installed in strict accordance with stress analysis reports and construction drawings, ensuring that their position and load settings are verified prior to hydrotesting and insulation.
- 4.13.4. Each support requires careful inspection to confirm alignment, load setting, and movement capability in both cold and hot conditions. Improper installation can result in hanger lock-up, excessive stress, or unanticipated load transfer to equipment nozzles. Sliding supports, spring hangers, and expansion joints must therefore be correctly oriented, locked during testing, and released under controlled conditions during commissioning.
- 4.13.5. Critical inspection and verification activities form an essential part of support installation. Material verification must confirm that support components such as steel structures, spring hangers, and snubbers conform to the specified grade and load rating.
- 4.13.6. Installation locations must be verified against stress analysis drawings, since even minor deviations can invalidate thermal movement calculations. Spring supports and snubbers must be adjusted with calibrated load cells or gauges, ensuring that cold and hot load settings are clearly marked for operational checks.
- 4.13.7. Welding of shoes, trunnions, or lugs must comply with approved WPS, and bolted connections must be tightened to the required torque. Flexibility must be verified to confirm that guides, anchors, and sliding supports allow intended movement without binding.
- 4.13.8. Finally, all adjustment points and inspection mechanisms must remain accessible after insulation is applied.
- 4.13.9. Rigid supports are typically used at anchors, guides, or rest points where restraint against

vertical or horizontal displacement is required. These supports directly transfer static and dynamic loads to the structural steel or foundation, thereby controlling the thermal expansion path. Installation involves welding or bolting support shoes or clamps to the pipe, ensuring full contact with the surface. Base plates are grouted and secured to the structure, with precise alignment confirmed by leveling instruments. Welding must follow approved procedures, and the final inspection must confirm that the support carries its design load without over-constraining the pipe.

- 4.13.10. Spring supports are applied where vertical thermal displacement must be accommodated while maintaining load transfer. Variable spring hangers are initially set to the cold load using calibrated adjustment bolts, with the travel scale locked until the system is pressurized. Constant spring hangers require careful alignment, with travel stops fixed until hydrotesting is completed. After the hydrotest and draining, and before hot operation, the stops are removed and the hot load positions are verified. Regular inspection ensures that springs operate within their design range and do not bind or misalign. These supports are essential in high-temperature lines such as steam piping.
- 4.13.11. Snubbers are installed to protect piping and equipment from sudden dynamic loads such as seismic events, water hammer, or turbine trips. While they restrain rapid displacement, they allow slow thermal movement. Hydraulic snubbers require air bleeding before service and must be verified for stroke length against design values. Mechanical snubbers must be aligned with manufacturer-specified clearances and tested before commissioning to confirm dynamic response. These devices are critical for protecting high-energy steam lines against shock and vibration.
- 4.13.12. Sliding supports provide controlled axial movement while restricting lateral displacement. They typically consist of a steel plate or PTFE pad, installed with precise alignment and lubrication to ensure smooth travel. Clearance must be checked against design tolerances to prevent binding while minimizing excessive play.
- 4.13.13. Special supports such as sliding shoes, sway braces, and struts are employed where additional control is required. Sliding shoes reduce friction along pipe racks, while sway braces and struts provide stability against wind, vibration, or seismic activity. Insulating pads such as PTFE, graphite, or composite plates are often installed to minimize wear and ensure unrestricted sliding.
- 4.13.14. All supports must be installed exactly as indicated in the approved drawings, with no field modifications permitted without engineering approval. Welding must follow qualified WPS, and surface preparation must be inspected before joining. After erection, alignment and elevation must be verified to ensure stresses remain within allowable limits.

- 4.13.15. Steam piping presents additional challenges due to significant thermal expansion and contraction. For this reason, spring supports must be adjusted to the cold setting during erection, with both cold and hot load positions marked for commissioning. Snubbers must be oriented correctly, function tested, and released before service. Insulation must be installed so as not to obstruct movement, with proper insulation saddles or clamps used to prevent pipe damage. Following hydrotest, spring supports must be reset, and snubber lock pins removed under the supervision of a piping support specialist.
- 4.13.16. To avoid construction delays, support installation should be scheduled concurrently with piping erection. For small-bore piping, supports may be directly installed by piping crews to improve efficiency. Proper planning, execution, and inspection of the support system are vital to ensuring the long-term reliability and safe operation of aboveground steam piping systems in combined cycle power plants.
- 4.13.17. After hydrostatic testing of steam piping, the cold setting of spring supports and snubbers must be adjusted. The support supplier typically dispatches a TA to verify compliance with design requirements and cold setting conditions, and issues a report requesting corrective actions for any deficiencies. Cold setting compliance must be strictly verified for all piping that will carry steam during the steam blowing period.

4.14. Pressure Test

- 4.14.1. Pressure testing represents a critical phase in piping construction, serving as the final verification of system strength, integrity, and leak tightness before commissioning. The default and most widely applied method is the hydrostatic test, in which liquid is used as the test medium.
- 4.14.2. Hydro testing is generally preferred because water stores far less energy than compressed gas and provides clear visual evidence of leaks, thereby minimizing stored-energy hazards. In certain situations where hydrostatic testing is impractical or would risk damaging sensitive equipment, pneumatic testing with gas may be employed; however, this method requires strict additional safety precautions because compressed gases contain significantly greater stored energy and present higher risks in the event of failure.
- 4.14.3. In limited cases where hydrostatic or pneumatic testing is impractical, an initial service leak test may be used with Owner approval. Under ASME B31.3 it is limited to Category D fluid service, and under ASME B31.1 (para 137.7) it may be applied to piping other than boiler external piping, which must always be pressure tested. These alternatives are permitted only where conventional testing cannot reasonably be performed, and their acceptance depends on explicit authorization.
- 4.14.4. The governing standards for these activities are ASME B31.1 (Power Piping) and ASME B31.3

(Process Piping), and both codes are applied to the systems designated to each. B31.1 applies to power piping such as main, reheat, and bypass steam, feedwater, condensate, and steam drain systems, including boiler external piping. B31.3 applies to systems designated as process piping by the project specification, such as fuel gas, fuel oil treatment, and chemical dosing systems, and provides detailed requirements for both hydrostatic and pneumatic testing, including formulas for determining test pressures, stepwise pressurization methods, and exemptions for specific components. Each test package must identify the governing code and the corresponding test pressure, hold time, and test medium.

- 4.14.5. ASME B31.1 (Power Piping), which applies to boiler-external piping and power plant systems, mandates hydrostatic testing as the standard method and specifies minimum hold times and strict procedural requirements. Where pressure vessels, boilers, or other specialized components are included within the test boundary, the ASME Boiler and Pressure Vessel Code (BPVC) and vendor-specific test requirements also apply, ensuring that vessel limitations are not exceeded.
- 4.14.6. Project Technical Specifications and the Owner's Purchase Orders take precedence over default code requirements and may impose stricter or modified conditions on test medium, pressure, or duration. These documents must always be reviewed and issued as the final governing requirements before any test is undertaken.
- 4.14.7. The hydrostatic test procedure begins with preparation of a detailed, approved written method statement. This procedure references the applicable code, test pressure calculations, selection of test medium, venting and draining arrangements, isolation strategy for non-tested equipment, and emergency response plans. Mechanical completion of the test boundary must be verified in advance, including the review of NDE results, PWHT certificates, material traceability, and support or hanger settings.
- 4.14.8. All non-rated items within the test section must be removed or isolated, and proper vents must be installed at high points to eliminate air pockets. The system is then filled slowly with clean, non-corrosive water (or other approved liquid), ensuring air is purged completely to avoid masking leaks or causing water hammer during pressurization. Certified and calibrated pressure gauges, recorders, pumps, hoses, and fittings must be installed and checked to confirm their suitability for the intended test pressure.
- 4.14.9. Pressurization is carried out in gradual increments, typically beginning at half of the intended test pressure or as otherwise specified by the governing code. At each stage, time is allowed for strains to equalize and for visual inspection of joints and welds before proceeding.
- 4.14.10. Full test pressure is maintained for the duration prescribed by code or specification, during which inspectors perform a detailed leak examination and record pressure-time data. Upon completion, pressure is reduced in a controlled manner, and if required by code, the system

may be held at design pressure for further sensitive leak detection using bubble or tracer-gas methods. Results, including any repairs and retests, are documented in the final test report.

- 4.14.11. When pneumatic testing is unavoidable, additional safeguards are mandatory. A formal risk assessment must be performed, and a written method statement prepared. Relief devices set below the maximum allowable overpressure must be installed.
- 4.14.12. Pressure application must be gradual, conducted in small steps, and all nonessential personnel must be excluded from the hazard zone. Where practical, barriers and shields are installed to contain the risk. Leak detection methods such as bubble tests, soap solution, or tracer-gas mass spectrometer analysis are applied only after the pressure has been reduced to the designated hold level. Due to the inherent risks, inert, non-flammable gases such as nitrogen are recommended as the test medium.
- 4.14.13. All pressure testing requires the use of calibrated instruments traceable to national standards. A pressure-time chart must be maintained, supplemented by photographic or video evidence of critical joints and test assemblies.
- 4.14.14. QA/QC or the Owner, must be appointed to oversee the process and sign the final test certificate. Test logs must record pressure increments, hold durations, temperature conditions, and any leaks identified with associated corrective actions.
- 4.14.15. In addition, a Permit-to-Work must be issued, safety watchers appointed, exclusion zones established, and all temporary test equipment verified for suitability.
- 4.14.16. Hydrostatic test pressure is not less than 1.5 times the design pressure under both ASME B31.1 and ASME B31.3. Under ASME B31.3, the test pressure is corrected for the difference between the test temperature and the design temperature using the allowable stress ratio (ST/S , not to exceed 6.5), and under both codes it must not exceed the maximum allowable test pressure of any non-isolated component.
- 4.14.17. Pneumatic tests, when permitted, are performed at 1.1 times the design pressure under ASME B31.3 and at not less than 1.2 and not more than 1.5 times the design pressure under ASME B31.1, and the pressure must not exceed the maximum allowable test pressure of any non-isolated component.
- 4.14.18. The pneumatic test procedure follows the stepwise method of the applicable code. The pressure is increased gradually to the lesser of one half of the test pressure or 170 kPa (25 psi) for a preliminary check of all joints, then increased in steps, holding at each step to equalize piping strains, until the test pressure is reached, and finally reduced to the design pressure before examination for leakage.
- 4.14.19. Where specified for ASME B31.3 systems such as fuel gas, a sensitive leak test is performed

by the bubble test or by an alternative method of demonstrated sensitivity, at a pressure not less than the lesser of 105 kPa (15 psi) gauge or 25% of the design pressure.

- 4.14.20. Under both codes the hydrostatic test pressure is held for a minimum continuous period of 10 minutes before the joints are examined, unless longer durations are required by the contract or the project specification. Expansion joints and bellows are treated as special cases, with temporary restraints or adjusted pressures applied if manufacturer limits are lower than piping test pressures.
- 4.14.21. High-pressure hydrotests impose additional requirements. All welds must have completed the necessary NDE examinations and any required PWHT prior to testing. Attached equipment must be verified for rating compatibility, with vessels isolated or otherwise accommodated if their limits are below the piping test pressure.
- 4.14.22. Spring supports, hangers, and snubbers must be set to cold positions, and expansion-joint thrust loads must be addressed with temporary restraints.
- 4.14.23. Pre-test checks confirm the torque of blind flanges, spool bolts, and temporary assemblies, as well as material suitability at the planned test temperature. OEM approval is required when integrated tests involve turbines or HRSG internals.
- 4.14.24. During testing, inspectors must monitor pressure and temperature continuously. Any unexplained pressure drop or visible leak requires immediate depressurization, repair, and retest. Acceptance criteria include the absence of visible leakage and a documented hold at the required pressure.
- 4.14.25. After completion, the system is depressurized gradually, drained, and dried, with further cleaning or insulation applied as necessary.
- 4.14.26. Test results, supporting data, and signatures from supervisors and inspectors are compiled into the QA record for permanent retention.
- 4.14.27. In practice, hydrostatic testing is always preferred where possible, due to its lower risk profile and straightforward leak detection.
- 4.14.28. Pneumatic testing is reserved for cases where liquid testing cannot be performed, and then only with robust risk mitigation, including shielding, remote monitoring, and safety barriers.
- 4.14.29. Any extended or integrated hydrotest that includes rotating equipment must be coordinated with the equipment OEMs and commissioning authorities to avoid damage or warranty conflicts. By adhering to these requirements, pressure testing not only validates piping integrity but also ensures safe transition from construction to operation.

4.15. Others

4.15.1. Control Valve

- 4.15.1.1. When a Control Valve is supplied, the Hydro Test period must be coordinated with the site visit of the valve supplier's TA.
- 4.15.1.2. Prior to the Hydro Test, remove the Internal Parts and the actuator from the Control Valve, and install a Blind Flange on the top face of the Valve Body.
- 4.15.1.3. After the Hydro Test is completed, re-install the Internal Parts and the actuator under the witness of the valve supplier's TA.
- 4.15.2. Check Valve
 - 4.15.2.1. Check valves must be installed in the flow direction indicated by the arrow on the valve body, and the direction must be confirmed against the P&ID and ISO drawings before installation.
 - 4.15.2.2. Installation orientation must follow the manufacturer's requirements. Swing check valves are generally installed in horizontal lines with the hinge pin horizontal and the cover facing up, or in vertical lines with upward flow only, with adequate straight pipe run as recommended by the manufacturer to prevent disc flutter.
 - 4.15.2.3. For hydro testing, check valves that would block filling, venting, or reverse-direction test pressure, or whose internals cannot withstand the test condition, must have the disc and internals removed or gagged open in accordance with the manufacturer's instructions, or be replaced with a temporary spool, and restored after the test. At stop valve locations in the HRSG steam circuits, check valves are temporarily removed and restored as described in the HRSG Hydro Test section.
 - 4.15.2.4. Before flushing, chemical cleaning, and steam blowing, check valve internals and soft seats are removed or protected to prevent damage from debris, and are reinstalled after cleanliness is confirmed.
- 4.15.3. For piping connected to instrument tubing lines, Hook-Up Drawings must be reviewed as needed to ensure that the instrumentation accurately reflects the fluid characteristics. For steam and other condensable services, instrument taps are taken from the side of the process pipe, and impulse lines are sloped with condensate pots or seal legs so that the sensing line remains filled with condensate. For liquid services such as fuel oil and water, taps are taken from the side or slightly below the horizontal centerline to prevent gas entrapment, and for air and gas services, taps are taken from the top of the pipe to prevent liquid accumulation. In all cases, the design requirements specified by the instrument vendor and the Hook-Up Drawings must be verified and followed.
- 4.15.4. Piping connected to the suction and discharge of pumps must strictly comply with spacing and straight-run requirements. Typically, a minimum straight length of 5D (five times the nominal pipe diameter) is maintained upstream of the pump suction nozzle to ensure a uniform flow

profile and prevent turbulence that can affect pump performance, unless the pump vendor specifies otherwise.

4.15.5. For piping connected to various control valves, including steam bypass control valves, hydro testing must be completed before commissioning. During startup and functional testing, the TA will usually visit the site to reinstall internal valve parts and actuators. Therefore, hydrotest schedules must be coordinated with the TA visit to ensure proper commissioning sequence.

5. Lessons Learned

5.1. Subject: Modification of HP Steam system supports due to heavy rubbing in HP Turbine

5.1.1. Issue

5.1.1.1. During the Performance Test after ST synchronization, sudden high vibration occurred. Investigation revealed heavy rubbing inside the HP Turbine, where the rotor came into contact with the diaphragm, causing severe vibration.

5.1.2. RCA (Root Cause Analysis)

5.1.2.1. Heavy rubbing inside the HP Turbine can occur due to multiple factors. The turbine supplier conducted a root cause analysis and identified potential issues:

5.1.2.2. Official Root Cause

- Thermal expansion of the HP Steam piping was constrained at the HP Turbine Main Stop Valve (MSV) connection due to improper support design and missing factors in stress analysis.

5.1.2.3. Potential Root Cause

- Thermal Expansion Issue: Incorrect thermal expansion calculations or violation of pre-warming procedures can cause insufficient clearances, leading to contact. Rapid load changes or cold starts increase this risk.
- Misalignment: Improper alignment between the shaft, bearings, or rotating components due to poor installation, foundation settlement, or thermal deformation.
- Rotor Bowing/Eccentricity: Uneven heating causes rotor deformation, resulting in contact with stator parts.
- Foreign Object Damage (FOD): Debris, loose parts, or broken components entering the steam path can damage blades and stationary parts, reducing clearances.
- Bearing Failure or Poor Support: Worn bearings or lubrication failure leading to shaft displacement or thrust bearing failure causing axial rotor shift.
- Blade Damage/Deformation: Fatigue, creep, or erosion may deform blades, reducing clearances.

- Design/Manufacturing Defects: Inadequate clearance design or poor tolerances.

5.1.3. Corrective Actions

- 5.1.3.1. Disassembly, repair, and reinstallation of the HP Turbine.
- 5.1.3.2. Redesign and modification of HP Steam piping supports to accommodate turbine movement.
- 5.1.3.3. Improvement of operational logic during commissioning

5.1.4. Preventive Measures

- 5.1.4.1. Install HP Steam piping from the turbine outward to avoid cold pulling.
- 5.1.4.2. Ensure piping supports comply strictly with drawings, including gap and cold-setting requirements.
- 5.1.4.3. Apply rigorous review of turbine connection piping to guarantee freedom of thermal expansion.

5.2. Subject: Non-compliance with steam piping slope requirements

5.2.1. Issue

- 5.2.1.1. During operation, water hammering risk was identified due to non-compliance with slope requirements in steam piping between HRSG and the Steam Turbine. Condensate water accumulation occurred where slope was inadequate or reversed.

5.2.2. RCA (Root Cause Analysis)

- 5.2.2.1. Steam piping must be installed with continuous slope towards drains to remove condensate. However, spools were installed with elevation mismatches, leading to reverse slope conditions. Fit-up inspections failed to check slope compliance.

5.2.3. Corrective Actions

- 5.2.3.1. Additional drain lines were installed downstream of CRH valves.
- 5.2.3.2. Drain points were added at locations with improper slope or reverse slope.

5.2.4. Preventive Measures

- 5.2.4.1. Strict compliance with slope requirements during installation and fit-up inspections.
- 5.2.4.2. Verification of slope against design drawings before final weld-up.

5.3. Subject: Improper Installation of Steam Piping Supports

5.3.1. Issue

- 5.3.1.1. Steam piping supports were installed without adhering to drawing requirements, resulting in inadequate thermal expansion control during commissioning.

5.3.2. RCA (Root Cause Analysis)

5.3.2.1. Support design requirements based on stress analysis were not followed due to poor workmanship and insufficient understanding of drawings. To meet schedule, supports were hastily installed without proper inspection, leading to malfunction during hot function tests.

5.3.3. Corrective Actions

5.3.3.1. Vendor technical advisors inspected installation and cold settings, and supports were modified or reinstalled accordingly.

5.3.4. Preventive Measures

5.3.4.1. Assign experienced support installation teams for steam systems.

5.3.4.2. Allocate sufficient time for installation of large-bore rigid, spring, and snubber supports.

5.3.4.3. Enforce strict inspection of cold setting and gap conditions before hot commissioning.

5.4. Subject: Incorrect Installation Orientation of Large-Diameter Butterfly Valve

5.4.1. Issue

5.4.1.1. A DN900 butterfly valve in the Circulation Water system was installed in the opposite direction due to design drawing error.

5.4.2. RCA (Root Cause Analysis)

5.4.2.1. Valve was installed according to ISO drawings, but flow direction marking conflicted with actual design intent.

5.4.3. Corrective Actions

5.4.3.1. Corrected valve installation direction in coordination with the design team.

5.4.4. Preventive Measures

5.4.4.1. Verify flow direction consistency between ISO drawings and P&IDs prior to installation.

5.5. Subject: Excessive Piping Movement due to Missing Expansion Joint Tie-Rods

5.5.1. Issue

5.5.1.1. During pump startup, excessive movement occurred in MCWP and ACWP GRP piping due to missing tie-rods at expansion joints.

5.5.2. RCA (Root Cause Analysis)

5.5.2.1. Tie-rods were omitted, leaving expansion joints free to move under pressure, causing displacement of buried GRP piping.

5.5.3. Corrective Actions

5.5.3.1. Installed tie-rods to restrain expansion joints.

5.5.4. Preventive Measures

5.5.4.1. Ensure expansion joints are installed with complete tie-rod assemblies as per design.

5.6. Subject: Incorrect Elevation of Steam Drain Piping (Design Error)

5.6.1. Issue

5.6.1.1. Steam drain piping was installed at a higher elevation than the atmospheric flash tank and vacuum tank, preventing condensate drainage.

5.6.2. RCA (Root Cause Analysis)

5.6.2.1. Design oversight: drain piping must always be lower than the steam header to allow gravity drainage. Installation was performed per flawed drawings.

5.6.3. Corrective Actions

5.6.3.1. Modified piping route to correct elevation.

5.6.4. Preventive Measures

5.6.4.1. Review elevation requirements during design to ensure drain piping is below the header.

5.7. Subject: Sampling System Piping Blockage and Tie-in Mismatch

5.7.1. Issue

5.7.1.1. Sampling lines between HRSG and Sampling Rack were mismatched at tie-in points, and some lines were blocked due to unremoved plastic end caps.

5.7.2. RCA (Root Cause Analysis)

5.7.2.1. Incorrect routing due to inadequate verification of HRSG nozzle and sampling rack nozzle locations.

5.7.2.2. In one case, plastic end caps were not removed prior to socket welding, blocking the line.

5.7.3. Corrective Actions

5.7.3.1. Reinstallation of incorrectly routed lines.

5.7.3.2. Removal of blockages through cutting, blowing, and cooler replacement.

5.7.4. Preventive Measures

5.7.4.1. Verify routing through 3D model review before construction.

5.7.4.2. Ensure end cap removal and internal cleanliness checks during fit-up inspections.

5.8. Subject: Omission of GT Fuel Oil Filter

5.8.1. Issue

5.8.1.1. Fuel oil filter required for turbine protection was omitted from the GT fuel oil supply system, requiring reinstallation.

5.8.2. RCA (Root Cause Analysis)

5.8.2.1. Design oversight during fuel oil system specification. Filter should have been included by the equipment supplier but was missed.

5.8.3. Corrective Actions

5.8.3.1. Installed final filter at GT fuel oil skid inlet with piping modifications.

5.8.4. Preventive Measures

5.8.4.1. Confirm fuel system requirements for GTs and ensure filters are included upstream of fuel oil/gas supply before construction.

5.9. Subject: CCW Supply & Return Piping Branch Line Isolation Valve

5.9.1. Issue

5.9.1.1. CCW system supplies cooling water to GT & ST coolers, heat exchangers, and pumps. However, branch lines were designed without isolation valves. As a result, once the header piping was installed, every time a hydro test was required for a new branch connection, the entire CCW system had to be shut down and repeatedly tested. This caused commissioning schedule delays and unnecessary repetitive testing.

5.9.2. RCA (Root Cause Analysis)

5.9.2.1. The design team applied plant operational criteria only, installing isolation valves only at the equipment inlets.

5.9.2.2. In practice, CCW piping and supports must be installed sequentially depending on building and equipment availability. Without branch isolation, hydro testing after each partial installation required full system shutdown and retesting.

5.9.2.3. In particular, the Steam Turbine side coolers and heat exchangers were installed later, after the Gas Turbine commissioning schedule had already begun. This created conflicts since hydro testing of new CCW branches required halting ongoing operations.

5.9.2.4. From a design perspective, isolation valves were omitted because they were not functionally necessary during plant operation and would add cost and accessibility issues.

5.9.3. Corrective Actions

5.9.3.1. Where required, isolation valves were installed on branch lines at site.

5.9.3.2. Alternatively, commissioning schedules were coordinated so that hydro tests requiring system shutdown could be performed with minimal operational impact.

5.9.4. Preventive Measures

5.9.4.1. Future designs should consider construction and commissioning sequencing.

5.9.4.2. Where practical, include isolation valves on CCW branch lines to minimize repeated system shutdowns and redundant testing.